Heterogeneous Prototype Distillation With Support-Query Correlative Guidance for Few-Shot Remote Sensing Scene Classification

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***Abstract*— Few-shot remote sensing scene classification (FSRSSC) aims to identify unseen classes only relying on very limited training samples. However, scarce training samples are insufficient to support a robust classwise representation, which is easily influenced by agnostic biases from diverse testing scenarios. Fortunately, there are abundant spatial contextual clues that exist in very limited training samples to have an enormous potential to establish discriminative and transferable concepts. Thus, in this article, a hybrid architecture called ProtoConViT is proposed to learn a powerful classwise representation based on spatial contextual clues for FSRSSC promotion. First, support-query correlative guidance is designed to generate more stable spatial connections among support and query data based on intermediate convolution neural network (CNN) feature maps, which not only can be embedded into each episodic training task to reduce redundant spatial contextual representation learning space of vision transformer (ViT) but also can assist it in rapidly capturing critical spatial contextual clues to classify query data into one of classes from support set. Second, followed by the designed support-query correlative guidance, a novel heterogeneous prototype distillation is proposed to integrate the advantages of CNN and ViT for heterogeneous prototype construction, which can rapidly set up discriminative and transferable concepts for FSRSSC. Third, corresponding to the proposed ProtoConViT, a joint loss is designed to make the model rapid convergence based on meta-learning. Finally, extensive experiments are carried out on three FSRSSC benchmarks, and comparative results indicate that the proposed ProtoConViT can achieve a superior FSRSSC performance.**

***Index Terms*— Convolution neural network (CNN), few-shot remote sensing scene classification (FSRSSC), heterogeneous prototype distillation, support-query correlative guidance, vision transformer (ViT).**

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I. Introduction

CLASSICAL remote sensing scene classification (RSSC) methods have to manually label sufficient data for model training, and meanwhile, they can set up an effective representation to work well for known class identification. However, different from several classical RSSC methods [1], [2], [3], [4], [5], under practical application scenarios, only scarce target samples can be acquired to support the model training process. In addition, referring to practical application demands, it is also required to build a more robust representation to identify valuable unseen classes in diverse testing scenarios. Naturally, few-shot RSSC (FSRSSC) is gradually becoming a challenging task. In view of this, the advanced concepts team has organized an FSRSSC competition based on the OPS-SAT case [6]. Their main purpose is to explore the few-shot learning ability for artificial intelligence onboard because it can sufficiently release burdens of data collection, relabeling, and model retraining to accelerate their launch program. At the same time, the few-shot learning ability can make satellites self-deal with much more unlabeled data related to unseen classes under real-time processing scenarios. It is obvious that the few-shot learning for RSSC should be the next-generation intelligent interpretation algorithm because it is more useful for practical applications such as military reconnaissance [7], scientific expedition [8], and resource exploration [9].

Currently, several studies [10], [11], [12], [13], [14] are focused on FSRSSC study. For example, Cheng et al. [10] designed a support-query prototype self-calibration mechanism to gracefully improve the prototype representation ability for few-shot learning according to the Siamese-prototype network. Tao et al. [11] and Chen et al. [12], respectively, introduced different pretraining pretexts and extra class name semantics at the model pretraining step for improving the FSRSSC performance. Li et al. [13] and Wang et al. [14] focused on learning the global–local and multiscale spatial information to advance FSRSSC ability. Notably, almost all previous FSRSSC studies [10], [11], [12], [13], [14] are constructed based on convolution neural network (CNN) because the inductive bias of CNN is very suitable for few-shot learning. As shown in Fig. 1(a), during stacked convolution feature extraction, different local spatial positions are designed

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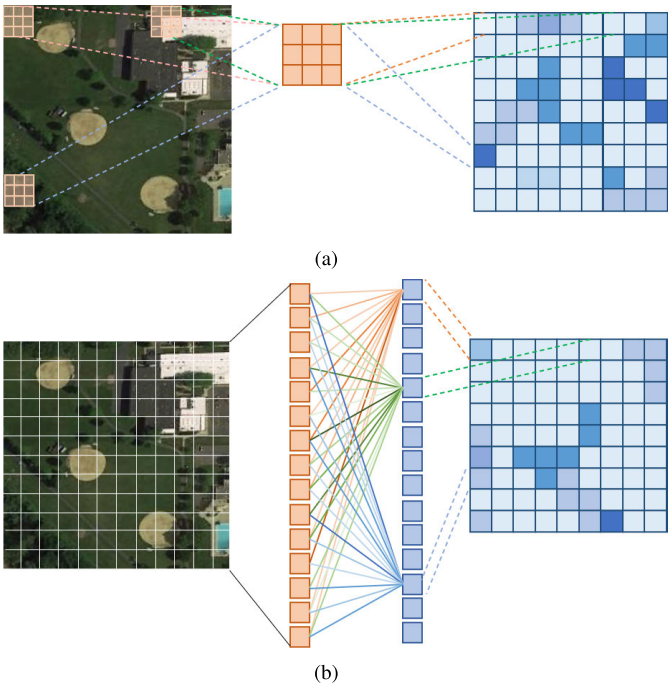


Fig. 1. Feature extraction diagrams of CNN and ViT. (a) Feature extraction way of sharing weight convolution for CNN. (b) Feature extraction way of multihead attention for ViT.

to share common convolution weights for spatial information aggregation, which would diminish a huge amount of learnable parameters and reduce much more training data requirement for discriminative representation learning. However, although CNN can rapidly abstract discriminative information from very limited training samples, a classwise representation based on spatial information aggregation (e.g., the stacked convolution and pooling operations) would inevitably ignore much more consistent spatial contextual clues between seen and unseen classes, which are not conducive to coping with agnostic biases from diverse testing scenarios of FSRSSC. Specifically, as illustrated in Fig. 2(a), under very low-data resource regimes (i.e., five-way one shot or five shots), CNN can effectively learn five classwise representations (i.e., black hollow circles) preparing for FSRSSC, but, owing to insufficient intraclass information in scarce training samples, more scattering feature distributions (i.e., blue solid circles) would be generated, and they are easily confused by agnostic biases for unseen class identification (i.e., green solid circles). Accordingly, although a lot of efforts [10], [11], [12], [13], [14] have been done on FSRSSC, they only can bring marginal improvements.

Except for building a classwise representation according to spatial information aggregation of CNN, due to more complex spatial layouts of remote sensing scenes, there are considerable spatial contextual clues in very limited training samples, which have enormous potential but are underexplored for FSRSSC. Currently, there are a lot of few-shot learning studies [15], [16], [17], [18], [19], [20], [21] have been proved that to sufficiently explore spatial contextual clues for classwise representation is very benefit for adapting to the agnostic biases [22] from diverse few-shot classification scenarios. Therefore, several pioneering studies [23], [24], [25], [26], [27] are attempted to introduce vision transformer (ViT) for classwise representation based on spatial contextual clues to achieve few-shot classification because ViT has been exhibited a powerful spatial contextual modeling ability whether for nature scene [28] or classical RSSC tasks [29]. As shown in Fig. 1(b), critical spatial contextual clues can be seen as the long-term dependence among visual tokens (i.e., fragmentary spatial patches), which can be built by multihead attention mechanism. Unfortunately, although a lot of efforts [23], [24], [25], [26], [27] have been done on ViT to make it have a few-shot learning ability for rapidly capturing critical spatial contextual clues, it still exhibits very slow convergence speed and, thus, only can capture the scarce spatial contextual clues from redundant spatial information, owing to the absence of inductive bias and requiring to learn massive parameters. As illustrated in Fig. 2(b). Although these scarce spatial contextual clues definitely have a very strong transferability adapting to agnostic biases from diverse testing scenarios, they would also present poor discriminability and then result in inferior FSRSSC performance. Overall, CNN and ViT both have their own advantages and disadvantages for few-shot learning, and their advantages and disadvantages seem to be complementary to some extent. For this reason, under very low-data resource regimes (i.e., less than five shot samples per class), how to effectively integrate the advantages of CNN and ViT to make a model rapidly capture more critical spatial contextual clues and form transferable and discriminative concepts for few-shot classification is very significant for FSRSSC.

In this article, in order to explore abundant spatial information that existed in very limited training samples, and to set up a few-shot learning ability that can rapidly capture critical spatial contextual clues from abundant spatial information, a novel hybrid architecture called ProtoConViT is proposed to learn the classwise representation based on spatial contextual clues, which can rapidly set up transferable and discriminative concepts for FSRSSC promotion. First, before the metalearning step, the CNN and ViT branches in the proposed ProtoConViT are pretrained by different self-supervised learning ways. Second, support-query correlative guidance is proposed for the proposed ProtoConViT to generate more stable spatial connections among support and query data, and it can be embedded into each episodic training task to reduce the redundant spatial contextual representation learning space of ViT and assist in capturing more critical spatial contextual clues for classifying each query data into one of classes from support set. This can provide a few-shot learning ability, such as the human visual cognitive system (HVCS), which only requires to glimpse of a couple of samples and remember more critical spatial contextual clues to form transferable and discriminative concepts for unseen class identification. Third, aiming to rapidly set up transferable and discriminative concepts according to support-query correlative guidance, a heterogeneous prototype distillation is designed in the proposed ProtoConViT to integrate the advantages of CNN and ViT. At the same time, the inductive bias of CNN can be seamlessly injected into ViT, and then, it can make ViT capture the much more critical spatial contextual clues from

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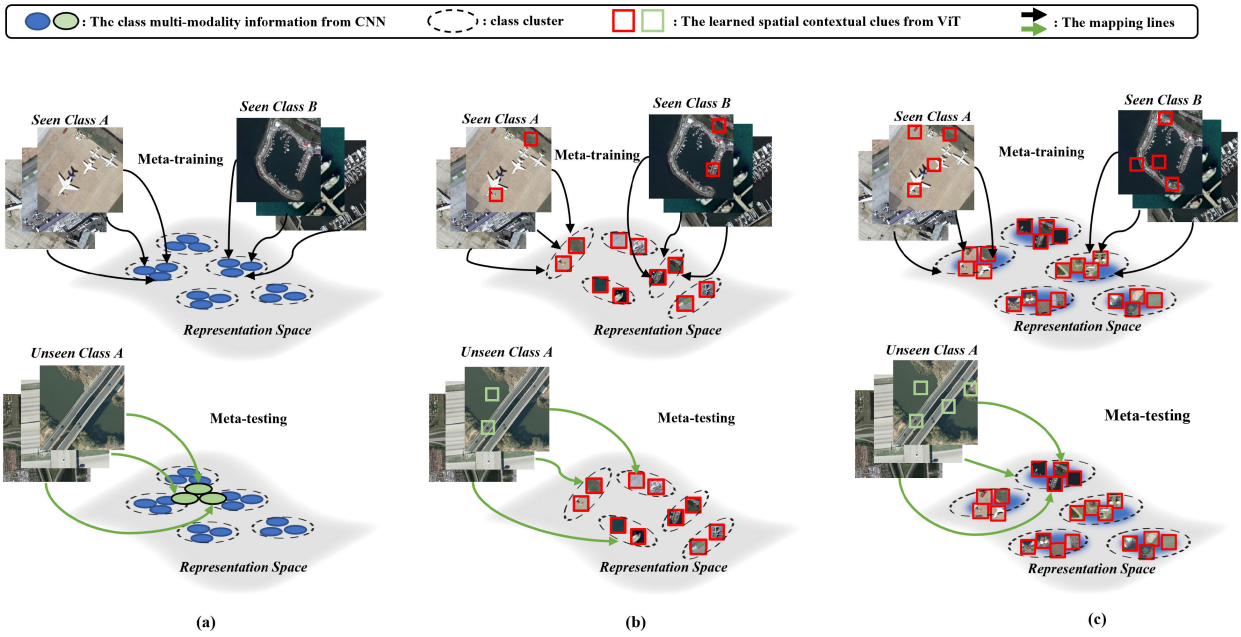


Fig. 2. Representation space diagram of different models. (a) Convolutional neural network. (b) ViT. (c) Hybrid architecture. The blue and green solid circles in (a) and (c) represent the learned class multimodality information from CNN. The red rectangles in (b) and (c) represent the learned spatial contextual clues from ViT. The blue solid circles and red rectangles represent prior-established feature representations at the training step. Green solid circles and rectangles represent feature representations at the testing step.

very limited training samples and boost forming transferable and discriminative concepts for FSRSSC by heterogeneous prototype construction. Accordingly, as shown in Fig. 2(c), a powerful heterogeneous prototype representation can be set up by the proposed ProtoConViT at training step, and then, it can provide transferable and discriminative concepts for well adapting to agnostic biases from diverse testing scenarios of FSRSSC. Fourth, corresponding to the proposed ProtoConViT, a joint loss is designed to make the model rapid and comfortable convergence. Finally, extensive experiments are carried out on several FSRSSC benchmarks including the datasets of AID [30], NWPU-RESISC45 [31], and UC Merced (UCM) Land-Use [32]. Under diverse few-shot learning scenarios, the results indicate that the proposed ProtoConViT can provide superior FSRSSC performance than state-of-the-art (SOTA) methods. In general, the main contributions can be summarized in the following.

1) Considering that there are abundant spatial contextual clues presented in very limited training samples and they are underexplored for FSRSSC, a novel hybrid architecture called ProtoConViT is proposed to effectively integrate the advantages of CNN and ViT, which can make the model rapidly learn critical spatial contextual clues and form transferable and discriminative concepts for achieving superior FSRSSC performance.
2) A support-query correlative guidance is designed to generate more stable spatial connections among support and query data, which not only can be embedded into each episodic training task for reducing the spatial contextual representation learning space but also can assist ViT in rapidly capturing more critical spatial contextual clues.
3) A novel heterogeneous prototype distillation is proposed to integrate the advantages of CNN and ViT, and it can build a heterogeneous prototype representation both having the discriminative multimodality information and transferable spatial contextual clues from CNN and ViT. Moreover, the inductive bias of CNN can be seamlessly injected into ViT, which can make ViT capture much more critical spatial contextual clues from very limited training samples and advance forming transferable and discriminative concepts for FSRSSC.

II. Related Work

A. Few-Shot Classification

Few-shot classification is a very important task in the computer vision field [33], which can achieve unseen class identification only relying on very limited training samples. Therefore, a lot of studies [34], [35], [36], [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [47], [48] are dedicated to set up a few-shot learning ability for image classification, and they can be divided into two categories including metalearning- [34], [35], [36], [37], [38], [39], [40], [41], [42], [43], [44] and transfer-learning-based [45], [46], [47], [48] [45], [46], [47], [48] methods.

First, according to metalearning-based methods [34], [35], [36], [37], [38], [39], [40], [41], [42], [43], [44], they are aiming to achieve the learning-to-learn according to episodic training tasks under very low-data resource regimes. Specifically, MatchingNet [34] and RelationNet [35] are proposed to establish the classwise similarity metric for support and query data under very low-data resource regimes, which can assist in achieving the learning-to-learn for unseen class identification. In addition, ProtoNet [36] is proposed to set up

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a valid prototype representation for few-shot leaning, which can be combined with a classwise similarity metric to learn a more suitable metric decision space. Subsequently, in view of abovementioned foundational studies [34], [35], [36], and in order to further improve the few-shot learning ability, Cheng et al. [37] proposed a specific feature disentangled representation method to make the model intend to learn more discriminative prototypes from very limited training samples. Xie et al. [38] designed a deep Brownian distance covariance metric to build a comprehensive classification space for few-shot classification promotion. Singh and Jamali-Rad [39] decoupled the representation as transferable semantic and discriminative label information to build a variational inference network, which can achieve unseen class identification under very low-data resource regimes. Except for setting up a more effective representation and classwise similarity metric space, there are several studies (e.g., MAML [40], Reptile [41], and MAML++ [42]) paying attention to parameter optimization for few-shot classification based on metalearning. Consequently, according to studies [40], [41], [42], Pandey and Yu [43] and Yao et al. [44] were dedicated to select and interpolate more effective episodic training tasks for model parameter optimization, which can upgrade few-shot learning efficiency and effectiveness based on metalearning.

Second, regarding transfer-learning-based methods [45], [46], [47], [48], the more effective model pretraining strategies are including to set up a representation with generalized knowledge and fine-tune it by a suitable adaptation way, which can alleviate data or class biases between pretraining and adaptation steps for unseen class identification under very low-data resource regimes. Concretely, Yazdanpanah et al. [45] proposed a feature distribution similarity measurement across stages of model pretraining and adaptation for few-shot classification. In addition, Li et al. [46], Chowdhury et al. [47], and Chen and Lee [48] were attempting to construct a universal and generalized representation by incorporating more diverse data or model knowledge at pretraining stage; therefore, they can achieve the few-shot classification by fine-tuning with a linear bias layer. Recently, MetaBaseline [49] and pretrain-metatrain-fine-tune (P > M > F) baseline [50] have considered metalearning as the suitable adaptation way to fine-tune the pretrained model, which can incorporate the transfer-learning with metalearning to achieve significant performance improvement for few-shot classification.

Therefore, in this article, inspired by the MetaBaseline [49] and (P > M > F) baseline [50], we are dedicated to establish a more suitable model pretraining way for hybrid architecture and study a heterogeneous prototype representation based on metalearning.

B. Spatial Information for Few-Shot Learning

As well as we know, only relying on very limited training samples makes it hard to set up an effective classwise representation, which would exhibit an undesirable few-shot classification performance. Fortunately, under very low-data resource regimes, there is abundant spatial information in very limited training samples and underexplored for few-shot learning. Therefore, recently, much more studies [23], [24], [25], [26], [27], [51], [52], [53], [54], [55] have already begun to pay much more attention to exploring the spatial contextual clues for classwise representation under very low-data resource regimes. For example, Chikontwe et al. [51], Lee et al. [52], and Zhang et al. [53] were focusing on setting up the transformer style self-attention mechanism among support and query data, which are aiming to improve the discriminability and transferability of local spatial feature representation via reweighting and cross-attention strategies. Consequently, Cao and Zhang [54] designed a local and global spatial mutual information maximization to set up a representation, having locally consistent across structural locations for improving few-shot learning ability. Afrasiyabi et al. [55] proposed a set of multiscale spatial structural representations by introducing the self-attention mechanism inside each scale feature layer, which can achieve better few-shot classification performance, depending on their designed set-to-set matching metric. Notably, the attention mechanism is widely used for exploring spatial contextual information, which can make the classwise representation to learn critical spatial structural or contextual clues from abundant spatial information and achieve impressive few-shot classification performance.

Since the multihead attention mechanism in ViT can set up a powerful spatial contextual representation according to fragmentary spatial visual tokens, several studies [23], [24], [25], [26], [27] have already begun to utilize ViT to explore the spatial contextual clues for few-shot classification. Lu et al. [23] proposed a dynamic hybrid ViT to enhance two inductive biases (i.e., the spatial relevance and diverse channel representation) based on CNN and ViT for few-shot classification. Dong et al. [24] extra introduced local-specific supervision to guide the multihead attention to learn which fragmentary spatial visual tokens are similar or dissimilar and, thus, accelerate the long-term dependence learning under very low-data resource regimes. Hiller et al. [25] proposed a spatial contextual few-shot learning way to help ViT rapidly build semantic correspondences between local spatial regions and independent of their respective class. Subsequently, referring to the framework of P > M > F [50], Xu et al. [26] designed a new attentive prefix tuning and domain residual adapter for ViT at model fine-tuning step, which can efficiently learn newly introduced tunable parameters over a novel support set. Then, Hao et al. [27] designed a class-aware patch embedding adaptation (CPEA) to learn the class-aware embeddings of image patches, which aims to integrate patch embeddings with class-aware embeddings for few-shot classification. Although a lot of effects [23], [24], [25], [26], [27] have been done on ViT to reduce the dependence of training data for massive parameter learning, there is still much room to be studied for improving few-shot learning ability of ViT.

In summary, whether for few-shot learning studies relying on CNN, ViT, or a hybrid architecture, they both demonstrated that the abundant spatial information that existed in very limited training samples has enormous potential to promote the few-shot learning ability. Accordingly, in this article, we are also aiming to continue exploring the spatial contextual information in very limited training samples for FSRSSC.

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C. Hybrid Architecture of CNN-ViT

Recently, there have been discussions on CNN or ViT, which is a better model that can provide a good enough representation in the computer vision field. In fact, CNN and ViT are respectively having their specific advantages for representation learning. At the same time, a lot of exploratory studies [5], [56], [57], [58], [59], [60], [61], [62], [63] are dedicated to rebuilding a hybrid architecture of CNN-ViT for more effective representation learning. For instance, Gao et al. [56] proposed a context aggregation network to integrate the inductive bias from CNN into long-term dependence of ViT, which would further improve the efficiency of representation learning. Peng et al. [57] proposed a conformer network to make the discriminative representation from CNN couple with the more transferable representation of ViT, which can further improve the image classification ability. Xie et al. [58] designed a more efficient ViT to reduce its massive data dependence for representation learning, where discriminative multimodality information from CNN was considered as prior information to accelerate more transferable spatial contextual clue construction from ViT. Then, Gao et al. [59] proposed a multiscale hybrid architecture of CNN-ViT incorporating a masked autoencoding strategy for model pretraining, and it can unleash the representation potential of hybrid architecture. Next, in order to avoid the lengthy large-scale pretraining of ViT, Zhang et al. [60] designed an agent CNN to introduce the inductive biases of CNN for ViT, which not only can improve the classification performance but also can advance model rapid convergence. Besides, Wang et al. [5], Deng et al. [61], Xu et al. [62], and Li et al. [63] developed a hybrid architecture of CNN-ViT for general classification tasks, and they are both focusing on designing a mixture module of CNN and ViT to accelerate spatial contextual representation learning.

In general, abovementioned studies [5], [56], [57], [58], [59], [60], [61], [62], [63] have been proved that hybrid architecture of CNN-ViT can exhibit a powerful representation learning ability for image classification, and they are both focused on integrating the inductive bias from CNN into ViT to assist model in rapid convergence or reducing massive training data dependence. Thus, in this article, we attempt to continuously explore the promising hybrid architecture of CNN-ViT for FSRSSC.

III. Methodology

In this section, the few-shot learning definition and hybrid architecture of the proposed ProtoConViT are introduced in Section III-A. The designed support-query correlative guidance and heterogeneous prototype distillation in the proposed ProtoConViT are elaborated on in Section III-B. Finally, corresponding to the proposed ProtoConViT, a joint loss is illustrated in Section III-C.

A. Overview

In order to achieve learning-to-learn for unseen class identification according to very limited training samples, there are training $\mathbb{D}_{\text{training}}$ and testing $\mathbb{D}_{\text{testing}}$ datasets for metalearning, consisting of a series of training tasks $\mathcal{T}_{\text{training}}$ and testing tasks $\mathcal{T}_{\text{testing}}$. Here, if $c(\cdot)$ represents the category of each sample whether in $\mathbb{D}_{\text{training}}$ or $\mathbb{D}_{\text{testing}}$ dataset, $c(\mathbb{D}_{\text{training}}) \bigcap c(\mathbb{D}_{\text{testing}}) = \emptyset$ is the necessary condition, which is aiming to project learnable knowledge from $\mathbb{D}_{\text{training}}$ into $\mathbb{D}_{\text{testing}}$ under very low-data resource regimes. In view of this, $\mathcal{T}_{\text{training}}$ would have the same setting as $\mathcal{T}_{\text{testing}}$ that can make the training process mimic testing procedure so-called episodic training tasks. In each training task $\mathcal{T}_{\text{training}}$, it is composed of support and query data for few-shot learning. Consequently, there are the support set $\mathcal{X}^s = \{(X_i^s, Y_i^s)\}_{i=1}^{n_s}$, i.e., $n_s = N \times K$ and query set $\mathcal{X}^q = \{(X_i^q, Y_i^q)\}_{i=1}^{n_q}$, i.e., $n_q = N \times q$ in $\mathcal{T}_{\text{training}}$ and $\mathcal{T}_{\text{testing}}$, where N represents the number of classes. K means the number of labeled samples per class in $\mathcal{X}^s$, which is usually to be set as a small number (e.g., from 1 to 5), and q is unlabeled samples per class in $\mathcal{X}^q$. Then, the goal of few-shot classification is to set up a more effective representation, which can classify the query sample X_i^q into one of N support classes only relying on K labeled samples from support set $\mathcal{X}^s$.

Therefore, followed by a metalearning framework, as shown in Fig. 3, a hybrid architecture called ProtoConViT is proposed to integrate the advantages of CNN and ViT for sufficiently exploring the spatial contextual information for FSRSSC. First, drawing inspiration from [11], [49], and [50], the model pretraining step should be considered before the metalearning step to build a solid foundation for few-shot learning. Accordingly, the CNN and ViT branches are individually pretrained by different self-supervised model pretraining ways because several studies [64], [65], [66] and our previous study [67] have been demonstrated that the self-supervised model pretraining based on unlabeled data can effectively construct the class-agnostic representation to avoid the overfitting phenomenon and possess a powerful few-shot learning ability. Specifically, MoCo [68] and MAE [69] are adopted for pretraining CNN and ViT branches in-domain because their specific pretexts can fully stimulate the ability of CNN and ViT to capture more discriminative multimodality information and transferable spatial contextual clues at model pretraining step. Second, based on pretrained CNN and ViT, and in order to explore abundant spatial information that existed in very limited training samples for improving the few-shot learning ability, support-query correlative guidance is designed for the proposed ProtoConViT, as shown in Fig. 3(a). It can be calculated from CNN feature maps of support and query data, and then, the more stable spatial connections among support and query data would be generated and embedded into each episodic training task, which not only can be used for reducing the redundant spatial contextual representation learning space and improve the few-shot learning ability of ViT but also can assist it in capturing the critical spatial contextual clues to classify the query sample X_i^q into one of N support classes only relying on scarce K labeled samples from support set $\mathcal{X}^s$. Subsequently, as shown in Fig. 3(b), followed by the designed support-query correlative guidance, a heterogeneous prototype distillation is designed to integrate the advantages of CNN and ViT for heterogeneous prototype construction. Thus, the discriminative multimodality information of CNN can be gracefully incorporated into transferable spatial contextual clues from ViT to generate the

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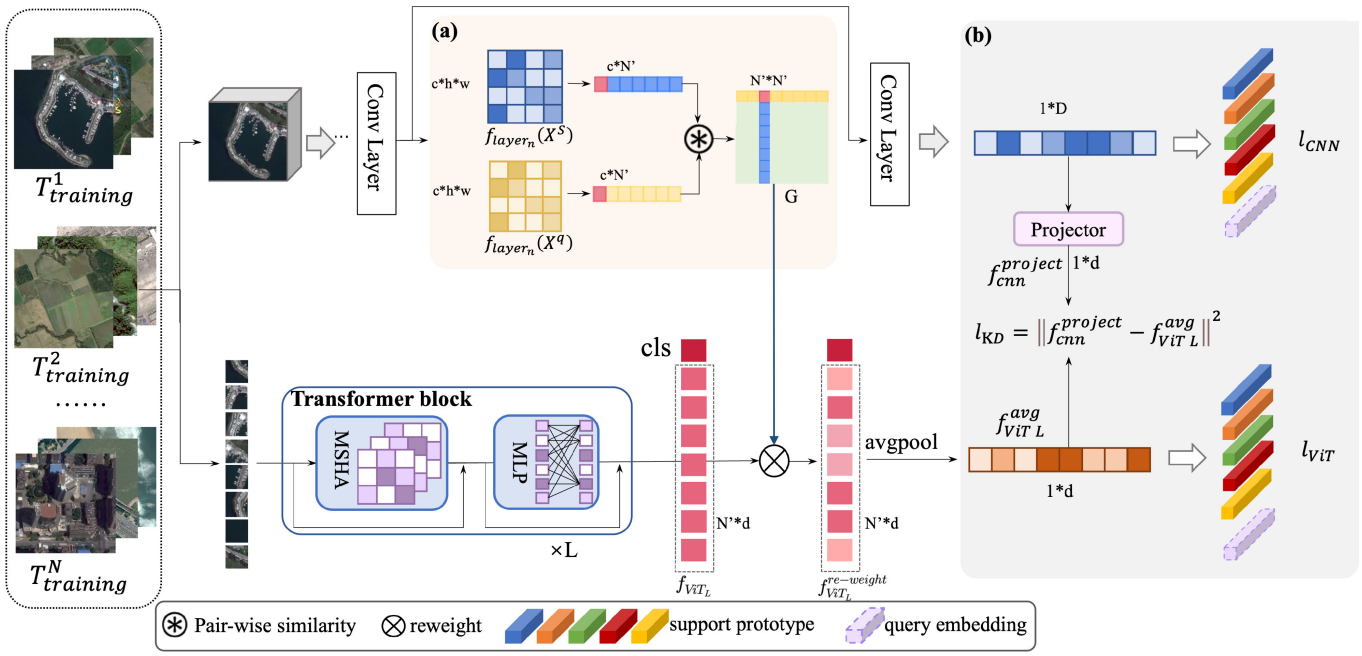


Fig. 3. Hybrid architecture of the proposed ProtoConViT at metatraining phase. (a) Support-query correlative guidance for reducing the redundant spatial contextual representation learning space. (b) Heterogeneous prototype distillation for integrating the advantages of CNN and ViT.

heterogeneous prototype representation, which can not only further improve the few-shot learning ability but also can introduce the inductive bias of CNN into heterogeneous prototype representation to encourage ViT capturing much more critical spatial contextual clues and rapidly form the transferable and discriminative concepts for FSRSSC. Third, corresponding to the hybrid architecture of the proposed ProtoConViT, a joint loss is designed to make the model rapid and comfortable convergence under very low-data resource regimes. In view of this, the designed heterogeneous distillation with support-query correlative guidance for the proposed ProtoConViT and corresponding joint loss is elaborated on as follows.

B. Hybrid Architecture of ProtoConViT

As advantages and disadvantages of CNN and ViT for few-shot learning are analyzed in Sections I and II, how to reasonably integrate abilities of the pretrained CNN and ViT to explore abundant spatial information presented in very limited training samples and improve the few-shot classification ability is very important for FSRSSC. Consequently, as shown in Fig. 3(a) and (b), the designed heterogeneous prototype distillation with support-query correlative guidance for the proposed ProtoConViT is aiming to rapidly explore abundant spatial contextual information and effectively integrate the advantages of CNN and ViT for FSRSSC. Next, more detail descriptions of the designed support-query correlative guidance and heterogeneous prototype distillation are in the following.

1) Support-Query Correlation Guidance: Aiming to further improve the few-shot learning ability of ViT in the hybrid architecture of the proposed ProtoConViT, and in order to assist in accelerating the more critical spatial contextual clues building-up from redundant local spatial information, a support-query correlation guidance is designed to reduce the representation learning space of ViT into an effective region. Thus, as shown in Fig. 3(a), when support and query data in each training task $\mathcal{T}_{\text{training}}$ are fed into the pretrained CNN and ViT, the intermediate feature maps of CNN from support and query data are used for support-query spatial correlation calculation, which can be expressed as follows:

$$G = \text{similar} < f_{\text{layer}_n}(\mathcal{X}^s) \cdot f_{\text{layer}_n}(\mathcal{X}^q) > . \tag{1}$$

In (1), similar $< \cdot >$ indicates the cosine similarity to calculate the point vector similarity among the intermediate CNN feature maps, e.g., $f_{\text{layer}_n}(\mathcal{X}^s) \in R^{c\times h\times w}$ and $f_{\text{layer}_n}(\mathcal{X}_q) \in R^{c\times h\times w}$, where, c, h, and w, respectively, represent the channel, width, and height information. The support-query spatial correlative guidance $G \in \mathbb{R}^{N'\times N'}$ can be obtained by (1), and it is set up based on the more stable spatial-modality information from intermediate CNN feature maps. $N' = w \times h$, and then, G can be seen as the guidance map to indicate which spatial regions are significantly connected between support and query data. Therefore, the generated G is considered to be embedded into each episodic training task to make ViT intend to learn more critical spatial contextual clues within support-query spatial correlative regions. Naturally, the spatial contextual representation learning space of ViT can be reduced, and the setup connections between support and query data can assist in classifying each query data into one of the classes from the support set only relying on a few training samples. Subsequently, the maximum operation is conducted on $G \in \mathbb{R}^{N'\times N'}$ along the vertical direction to generate the coefficients $G' \in \mathbb{R}^{N'\times 1}$ for reweighting the spatial contextual representation from ViT, which can be expressed as follows:

$$f_{\text{vit}_L} = \text{MLP}_L\left(\text{MSA}_L\left(\text{LN}_L\left(\left(x_p^1; \ldots; x_p^{N'}\right)E\right)\right)\right), \tag{2}$$

$$f_{\text{vit}_L}^{\text{re-weight}} = G' \otimes f_{\text{vit}_L}. \tag{3}$$

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In (2), f_{vit_L} indicates a spatial contextual representation from the Lth transformer block (i.e., containing the operations of layer normalization LN($\cdot$), multihead self-attention MSA($\cdot$), and multiple layer perception MLP($\cdot$)) based on a sequence of embedded fragmentary spatial visual tokens $(x_p^1; \ldots; x_p^{N'})E$. Next, in (3), $\otimes$ is the dot product, and then, it can be applied for $G' \in \mathbb{R}^{N' \times 1}$ and $f_{\text{vit}_L} \in \mathbb{R}^{N' \times d}$ to generate the reweighted spatial contextual representation $f_{\text{vit}_L}^{\text{re-weight}} \in \mathbb{R}^{N' \times d}$ in each episodic training task, which is preparing for heterogeneous prototype construction to form transferable and discriminative concepts for FSRSSC.

2) Heterogeneous Prototype Distillation: Under the designed support-query correlative guidance within each episodic training task, and aiming to effectively integrate the advantages of CNN and ViT to further improve the few-shot learning ability of hybrid architecture and form transferable and discriminative concepts for FSRSSC, the heterogeneous prototype distillation is designed to incorporate the multimodality representation $f_{\text{cnn}} \in R^{1 \times D}$ of CNN into the reweighted spatial contextual clues $f_{\text{vit}_L}^{\text{re-weight}} \in \mathbb{R}^{N' \times d}$ from ViT to achieve the heterogeneous prototype representation.

Then, as shown in Fig. 3(b), before the designed heterogeneous prototype distillation, $f_{\text{cnn}} \in R^{1 \times D}$ and $f_{\text{vit}_L}^{\text{re-weight}} \in \mathbb{R}^{N' \times d}$ should be projected into the same dimension, which can be expressed as follows:

$$f_{\text{cnn}}^{\text{project}} = \text{project}(f_{\text{cnn}}), \tag{4}$$

$$f_{\text{vit}_L}^{\text{avg}} = \text{AvgPool}\left(f_{\text{vit}_L}^{\text{re-weight}}\right). \tag{5}$$

In (4) and (5), project($\cdot$) and AvgPool($\cdot$) represent a learnable projecting function and average pooling operation, which individually make the multimodality and spatial contextual representations to have the same dimension $1 \times d$. Then, the heterogeneous prototypes of CNN and ViT can be, respectively, calculated by the following equations:

$$C_{\text{cnn},k} = \left(\frac{1}{S_k}\right) \sum f_{\text{cnn},k}, \tag{6}$$

$$C_{\text{vit},k} = \left(\frac{1}{S_k}\right) \sum f_{\text{vit}_L,k}^{\text{avg}}. \tag{7}$$

According to (6) and (7), $C_{\text{cnn},k}$ and $C_{\text{vit},k}$ are the calculated heterogeneous prototypes from CNN and ViT. S_k indicates a total number of kth labeled class samples from support set, and $k = 1, \ldots, N$. Since the heterogeneous prototypes can be set up by (6) and (7), the designed heterogeneous prototype distillation can be achieved by the following equation:

$$\{C_{\text{cnn}}, C_{\text{vit}}\} = \text{Dist}\left(f_{\text{cnn}}^{\text{project}}, f_{\text{vit}_L}^{\text{avg}}\right). \tag{8}$$

Followed by (8), $\{C_{\text{cnn}}, C_{\text{vit}}\}$ are the heterogeneous prototype representations, which can be generated by feature distillation Dist($\cdot$). Since Dist($\cdot$) is a soft way to integrate discriminative multimodality information from CNN into transferable spatial contextual clues of ViT, after heterogeneous prototype distillation, C_{vit} is considered as the heterogeneous prototype representation to possess both the powerful transferability and discriminability, and thus, it can well cope with diversity metatesting scenarios. In addition, as shown in Fig. 3(b), by the designed heterogeneous prototype distillation, the inductive bias of CNN is reasonably introduced into ViT, which not only can further improve few-shot learning ability to facilitate heterogeneous prototype construction but also encourage the proposed ProtoConViT to learn the much more critical spatial contextual clues for setting up transferable and discriminative concepts for FSRSSC.

C. Joint Loss for Hybrid Architecture

Corresponding to heterogeneous prototype distillation with support-query correlative guidance, a joint loss is designed for the hybrid architecture of the proposed ProtoConViT to assist the model in rapid and comfortable convergence based on metalearning. It can be written as

$$L_{\text{total}} = L_{\text{vit}} + \lambda L_{\text{cnn}} + L_{\text{KD}}. \tag{9}$$

In (9), L_{total} indicates the joint loss for the hybrid architecture of the proposed ProtoConViT. L_{vit} and L_{cnn} are the supervised prototype learning for ViT and CNN branches, respectively, and λ is the hyperparameter to balance the effect of multimodality information from CNN branch injecting into spatial contextual clues from ViT branch in a hybrid architecture. Here, L_{vit} and L_{cnn} can be expressed as

$$L_{\text{vit}} = -\sum_{k=1}^{N} \sum_{i=1}^{S_k} (y_i = k) \log\left(p_{\text{vit}}\left(\hat{y}_i = k | x_i^q\right)\right), \tag{10}$$

$$L_{\text{cnn}} = -\sum_{k=1}^{N} \sum_{i=1}^{S_k} (y_i = k) \log\left(p_{\text{cnn}}\left(\hat{y}_i = k | x_i^q\right)\right). \tag{11}$$

In (10) and (11), the cross-entropy loss is utilized for L_{vit} and L_{cnn}. Then, y_i is the sample category label information from support data, and $\hat{y}_i = k$ is a predicted label information for query data relying on $p_{\text{vit}}(\hat{y}_i = k | x_i^q)$ and $p_{\text{cnn}}(\hat{y}_i = k | x_i^q)$. Subsequently, $p_{\text{vit}}(\hat{y}_i = k | x_i^q)$ and $p_{\text{cnn}}(\hat{y}_i = k | x_i^q)$ can be calculated as follows:

$$p_{\text{vit}}\left(\hat{y}_i = k | x_i^q\right) = \frac{\exp\left(\text{dis}\left(f_{\text{vit}_L,i}^{\text{avg,query}}, C_{\text{vit},k}\right)\right)}{\sum_i \sum_k \exp\left(\text{dis}\left(f_{\text{vit}_L,i}^{\text{avg,query}}, C_{\text{vit},k}\right)\right)}, \tag{12}$$

$$p_{\text{cnn}}\left(\hat{y}_i = k | x_i^q\right) = \frac{\exp\left(\text{dis}\left(f_{\text{cnn},i}^{\text{query}}, C_{\text{cnn},k}\right)\right)}{\sum_i \sum_k \exp\left(\text{dis}\left(f_{\text{cnn},i}^{\text{query}}, C_{\text{cnn},k}\right)\right)}. \tag{13}$$

Followed by (12) and (13), $p_{\text{vit}}(\hat{y}_i = k | x_i^q)$ and $p_{\text{cnn}}(\hat{y}_i = k | x_i^q)$ can be calculated by cosine similarity measurement dis($\cdot$), which should compare each query feature representation $f_{\text{vit}_L,i}^{\text{avg,query}}$ or $f_{\text{cnn},i}^{\text{query}}$ with k class prototypes from support set according to equations of (6) and (7).

Besides, the third term L_{KD} in (9) represents the heterogeneous prototype distillation loss, which can be defined as

$$L_{\text{KD}} = \arg\min \left\| f_{\text{cnn}}^{\text{project}} - f_{\text{vit}_L}^{\text{avg}} \right\|_2. \tag{14}$$

Related to (14), the soft fusion of knowledge distillation can be achieved by L_{KD}. It can objectively evaluate the heterogeneous prototype distillation between $f_{\text{cnn},k}^{\text{project}}$ and $f_{\text{vit},k}^{\text{avg}}$ by similarity measurement, where the similarity measurement can be defined as the L_2 − norm based on $\| \cdot \|_2$ to pull the discriminative multimodality representation $f_{\text{cnn},k}^{\text{project}}$ and transferable spatial contextual clues $f_{\text{vit},k}^{\text{avg}}$ together. As a result,

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Algorithm 1 Hybrid Architecture of ProtoConViT

Input: The train dataset $\mathbb{D}_{training}$ containing episodic training tasks $\mathcal{T}_{traing}$ of support set $\mathcal{X}^s$ and query set $\mathcal{X}^q$; The pretrained CNN and ViT parameters θ_{CNN} and θ_{ViT}
Output: hybrid architecture parameter θ_{best}
for *each episode T* **do**
  Get the CNN feature $f_{cnn} \in R^{1\times D}$;
  Get the ViT feature $f_{vit_L} \in \mathbb{R}^{N'\times d}$;
  for *each pair of support and query data* **do**
    Obtain the intermediate CNN feature map $f_{layer_n}(\mathcal{X}^s)$ for support data;
    Obtain the intermediate CNN feature map $f_{layer_n}(\mathcal{X}^q)$ for query data;
    Calculate the support-query correlative guidance G by eq. (1);
    Transfer G into G' by maximum operation;
    By using G' to re-weight f_{vit_L} based on eq. (3);
  end
  Output $f_{vit_L}^{re-weight} \in \mathbb{R}^{N'\times d}$;
  Project f_{cnn} into $f_{cnn}^{project} \in R^{1\times d}$ by eq. (4);
  Project $f_{vit_L}^{re-weight}$ into $f_{vit_L}^{avg} \in R^{1\times d}$ by eq. (5);
  Calculate the prototype C_{cnn} by eq. (6);
  Calculate the prototype C_{vit} by eq. (7);
  Do distillation for $\{C_{cnn}, C_{vit}\}$ by eq. (8);
  Calculate the joint loss L_{total} by eq. (9);
  Update the parameters θ for hybrid architecture with back-propagation;
end
Return Parameter θ_{best};
% The heterogeneous prototype representation C_{vit} is used for meta-testing

the heterogeneous prototype distillation with support-query correlative guidance can significantly improve the few-shot learning ability in the hybrid architecture of the proposed ProtoConViT, and the completed training procedure is illustrated in Algorithm 1.

IV. Experiments

A. Dataset Description

The extensive experiments for FSRSSC are built and conducted on three available RSSC datasets, including NWPU-RESISC45 [31], AID [30], and UCM Land-Use [32], to verify the effectiveness for the hybrid architecture of the proposed ProtoConViT. Subsequently, three RSSC datasets with their corresponding metalearning-based FSRSSC settings are briefly described in the following:

1) NWPU-RESISC45: The RSSC dataset was created by Northwestern Polytechnical University, and it includes 45 remote sensing scene classes. Each class contains 700 images; thereby, there are a total of 31 500 images in the dataset. In addition, the image resolution is 256 × 256 pixels, and spatial resolutions vary from 0.2 to 30 m. For the experiment setting of FSRSSC, the classes in NWPU-RESISC45 are divided into 30 classes (i.e., seen classes) for metatraining and the remaining 15 classes (i.e., unseen classes) for metatesting. The details are listed in the following.

1) Seen Class in Metatraining: *Airport*, *airplane*, *baseball diamond*, *basketball court*, *beach*, *bridge*, *chaparral*, *church*, *circular farmland*, *cloud*, *dense residential*, *commercial area*, *desert, forest*, *freeway*, *golf course*, *ground track field*, *harbor*, *industrial area*, *intersection*, *island*, *meadow*, *lake*, *medium residential*, *mobile home park*, *mountain, overpass*, *palace*, and *parking lot*.
2) Unseen Class in Metatesting: *Railway station*, *rectangular farmland*, *river*, *roundabout*, *runway*, *sea ice*, *ship*, *snow berg*, *sparse residential*, *stadium*, *storage tank*, *tennis court*, *terrace*, *thermal power station*, and *wetland*.

2) AID: This is a large-scale dataset for aerial scene images, and all of the images are collected from Google Earth service, which contains 30 classes with a total of 10 000 images. Here, the number of images in each class varies from 200 to 400, and each image size is 256 × 256 pixels. In the FSRSSC experiment, 20 classes (i.e., seen classes) in AID are used for metatraining, and the rest of the ten classes (i.e., unseen classes) are utilized for metatesting. The details are listed in the following.

1) Seen Class in Metatraining: *Airport*, *bareland*, *baseball field*, *beach*, *bridge*, *church*, *commercial*, *dense residential*, *desert*, *forest*, *playground*, *industrial*, *meadow*, *center*, *medium residential*, *mountain*, *parking*, *park*, *pond*, and *farmland*.
2) Unseen Class in Metatesting: *Port*, *railway station*, *resort*, *river*, *school*, *square*, *sparse residential*, *stadium*, *storage tanks*, and *viaduct*.

3) UCM Land-Use: This dataset was created in 2010 at the University of California. All of the images were manually extracted from large images of the U.S. Geological Survey's National Map City Image Set. It contains 21 classes of scenes for cities and regions in the U.S., with 100 images for each type of scene, for a total of 2100 images. The image size is 256 × 256 pixels, and the spatial resolution is one foot. Related to the experiment of FSRSSC, we divided them into 11 classes (i.e., seen classes) of metatraining and ten classes (i.e., unseen classes) of metatesting. The details are listed in the following.

1) Seen Class in Metatraining: *Agricultural*, *airplane*, *baseball diamond*, *beach*, *buildings*, *chaparral*, *dense residential*, *forest*, *freeway*, *golf course*, and *harbor*.
2) Unseen Class in Metatesting: *Intersection*, *medium residential*, *mobile home park*, *overpass*, *parking lot*, *river*, *runway*, *sparse residential*, *storage tanks*, and *tennis court*.

It is necessary to highlight that our approach diverges from the typical setup in FSRSSC studies. There is no validation set because we are considering to increase the number of categories for metatesting. From our perspective, only relying on specific five or six classes to evaluate five-way classification performance can be overly fortuitous. Furthermore, in order to mitigate the potential variability in results, more than ten categories in metatesting scenarios are considered in the

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FSRSSC setting of NWPU-RESISC45 [31], AID [30], and UCM Land-Use [32] datasets.

B. Experimental Setting

Related to the hybrid architecture of the proposed ProtoConViT, three versions of ViT-B, ViT-S, and ViT-tiny [70] and two versions of Res-18 and Res-10 [71] are selected as the backbones (i.e., $hybrid$, $hybrid^{*}$, and $hybrid^{\dagger}$), which consist of Res-18 + ViT-B, Res-10 + ViT-S, and Res-10 + ViT-tiny, respectively. Then, for all versions of ViT, the final projector head was omitted to leverage the feature embedding; meanwhile, the final FC layer of all versions of CNN was also removed. Notably, it is worth noting that the backbones of CNN and ViT branches are not restricted, where any valid CNN and ViT backbones can be adopted for building hybrid architecture as long as the final representation dimensions are aligned.

At the stage of self-supervised model pretraining, the pretrained parameters from CSPT [67] are utilized as the initial class-agnostic representation for ViT. Then, since there are no available pretrained model parameters for Res-18 and Res-10, the self-supervised pretraining way of MoCo [68] is utilized for model pretraining with 200 epochs, and it can be treated as the initial class-agnostic representation for CNN. Related to the self-supervised model pretraining of CNN, we followed the default setting of MoCo, i.e., $\tau = 0.07$ and $K = 256$ corresponding to τ, k, and momentum $m = 0.999$. The learning rate was set to 0.001, and the SGD optimizer with a momentum of 0.9 was employed. In addition, considering the risk of overfitting under very low-data resource regimes and collapsing in diverse metatesting scenarios. Then, the maximum number of training epochs for metalearning was set as 20, which is the same as MetaBaseline [49]. Meanwhile, the limited epochs for metalearning are also requiring the model to rapidly acquire task-level knowledge to improve the few-shot learning performance on N-way K-shot tasks. During the metatraining process, the hybrid architecture of the proposed ProtoConViT was trained by the SGD optimizer, and the learning rate was empirically set to 0.001. Finally, 2000 tasks are randomly sampled for metatesting. For each task evaluation, the heterogeneous features $f_{\mathrm{vit}_L,i}^{\mathrm{avg,query}}$ from query set and heterogeneous prototype representation $C_{\mathrm{vit},k}$ from support set are measured by cosine similarity for final FSRSSC prediction, which can achieve N-way K-shot classification with 15 query samples per task. Finally, the performance was assessed based on average accuracy accompanied by 95% confidence intervals.

C. Comparison Analysis

In order to evaluate the effectiveness of the proposed ProtoConViT for FSRSSC, several classical and SOTA methods are selected as comparisons. Specifically, they can be summarized as three categories: 1) the classical few-shot learning way including the optimized and metric-based metalearning methods, e.g., the MAML [40], RelationNet [35], ProtoNet [36], and MatchingNet [34]; 2) the SOTA FSRSSC methods comprising the transfer-learning-based methods, e.g., IDSSL [11], CNSPN [12], and MES^{2}L [13], and metalearning-based methods, e.g., SPNet [10], MSFMNet [14]; and 3) the SOTA nature scene few-shot classification methods involving transfer-learning- and metalearning-based methods, e.g., the HelixFormer [53], SetFeat [55], CPEA [27], SUN [24], and FewTURE [25].

As reported in Tables I–III, a novel hybrid architecture of the proposed ProtoConViT can exhibit a superior FSRSSC performance whether for five-way one, two, three, four, or five shots on NWPU-RESISC45, AID, and UCM datasets than other comparisons, which is mainly attributed to more effective classwise representation based on spatial contextual clues for heterogeneous prototype construction. Subsequently, the comparison analysis is given as follows.

1) NMPU-RESISC45: As reported in Table I, the proposed ProtoConViT can achieve the 90.75% ± 0.33%, 94.33% ± 0.20%, 96.00% ± 0.15%, 96.37% ± 0.14%, and 97.16% ± 0.08% classification accuracies for 5w1s, 5w2s, 5w3s, 5w4s, and 5w5s, respectively, on NMPU-RESISC45. FewTURE [25] can achieve the second best performance, which is about 2%∼3% accuracy lower than the proposed ProtoConViT under all of the few-shot classification scenarios. Meanwhile, referring to the 14th and 15th lines in Table I, we can see that adopting unlabeled data in different domains for self-supervised pretraining would produce a significant impact on metalearning step for FewTURE [25]. Therefore, before the metalearning step, the self-supervised pretraining in-domain is necessary to achieve more competitive FSRSSC performance. In addition, considering that spatial contextual clues have the great potential for few-shot classification promotion, the SUN [24], FewTURE [25], and CPEA [27] are attempting to introduce the ViT for few-shot classification. Although SUN [24], FewTURE [25], and CPEA [27], respectively, designed a token important weight generator, specific local spatial region supervision, and CPEA to make ViT pay attention to learn critical spatial contextual clues from very limited training samples, the more complex spatial layouts of remote sensing scenes would make a considerable difficulty to generate more effective and stable token important weights, point out which specific local spatial regions should be emphatically learned, or find adaptive class-relevant image patches from very limited training samples. Naturally, the SUN [24], FewTURE [25], and CPEA [27] only can alleviate the issues (e.g., in the absence of inductive bias and massive learnable parameters with redundant representation space for searching critical spatial contextual clues) of ViT for few-shot learning to a certain extend and achieve not good enough FSRSSC performance.

Then, except for ViT-based few-shot classification methods, from the first line to the 11th line, we can see that the few-shot classification methods that are built upon CNN both perform lower FSRSSC performance because the feature extraction way of CNN would inevitably overlook a lot of spatial contextual information, which is not friendly for adapting to agnostic bias from diverse testing scenarios. To compare with classical few-shot classification methods (e.g., MAML [40], RelationNet [35], ProtoNet [36], and MatchingNet [34]), HelixFormer [53] and SetFeat [55] for

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TABLE I
FIVE-WAY CLASSIFICATION ACCURACIES (%) OF DIFFERENT METHODS ON THE NWPU-RESISC45 DATASET

Method	Publish	Backbone	Params	Flops	5w1s	5w2s	5w3s	5w4s	5w5s
MAML [40]	ICML(2017)	Conv4	0.13M	0.42G	47.81±0.44	63.12±0.40	67.71±0.36	70.25±0.36	76.14±0.47
RelationNet [35]	CVPR(2018)	Conv4	0.23M	0.39G	52.45±0.50	67.32±0.36	71.15±0.29	73.24±0.29	77.11±0.37
MatchingNet [34]	NIPS(2016)	Conv4	0.23M	0.40G	59.83±0.44	65.78±0.33	70.43±0.31	71.34±0.30	73.92±0.32
Protonet [36]	NIPS(2017)	Conv4	0.23M	0.40G	58.26±0.49	66.66±0.34	70.02±0.30	73.36±0.28	77.03±0.37
SPNet [10]	TGRS(2022)	Res-18	11.2M	1.82G	64.59±0.48	71.21±0.34	77.85±0.27	78.96±0.24	80.84±0.27
IDSSL [11]	GRSL(2021)	Res-18	11.2M	1.82G	55.66±0.22	66.16±0.39	71.83±0.36	74.86±0.34	76.80±0.16
HelixFormer [53]	ACM MM(2021)	Conv4	0.44M	12.87G	64.76±0.23	73.73±0.36	77.44±0.33	79.73±0.31	81.77±0.31
SetFeat [55]	CVPR(2022)	Resnet12	13.8M	4.89G	72.77±0.80	80.94±0.34	84.44±0.29	86.10±0.27	87.18±0.25
CNSPN [12]	TGRS(2022)	Res-18	13.96M	1.83G	69.96±0.41	75.37±0.36	78.67±0.32	80.14±0.30	80.79±0.30
MES^{2}L [13]	TGRS(2023)	Resnet12	12.9M	13.45G	61.99±0.82	70.79±0.38	75.29±0.33	78.41±0.31	81.23±0.50
MSFMNet [14]	TGRS(2024)	Resnet12	8.6M	3.12G	74.67±0.42	82.38±0.32	85.60±0.28	87.29±0.24	88.30±0.22
CPEA [27]	ICCV(2023)	ViT-B(CSPT)	95.8M	16.90G	66.68±0.50	80.10±0.37	84.11±0.32	87.29±0.26	88.38±0.24
SUN [24]	NIPS(2022)	Visformer	12.5M	1.00G	75.80±0.42	82.75±0.33	85.88±0.28	87.72±0.26	89.17±0.23
FEWTURE [25]	NIPS(2022)	ViT-B	86.2M	16.86G	72.63±0.37	89.09±0.35	83.38±0.28	85.44±0.27	86.79±0.24
FEWTURE [25]	NIPS(2022)	ViT-B (CSPT)	86.2M	16.86G	86.82±0.67	91.50±0.25	92.93±0.21	93.70±0.18	95.48±0.38
ProtoConViT	ours	$hybrid$	98.1M	18.68G	**90.75±0.33**	**94.33±0.20**	**96.00±0.15**	**96.37±0.14**	**97.16±0.08**
		$hybrid^*$	26.3M	5.14G	77.94±0.43	88.26±0.28	90.72±0.22	92.01±0.20	92.74±0.17
		$hybrid^\dagger$	10.5M	1.97G	76.42±0.44	85.06±0.32	87.79±0.26	89.21±0.24	90.39±0.22

The $hybrid$, $hybrid^*$ and $hybrid^\dagger$ represent Res-18+ViT-B, Res-10+ViT-S and Res-10+ViT-tiny. The best and second best results are bold and underline.

TABLE II
FIVE-WAY CLASSIFICATION ACCURACIES (%) OF DIFFERENT METHODS ON THE AID DATASET

Method	Publish	Backbone	Params	Flops	5w1s	5w2s	5w3s	5w4s	5w5s
MAML [40]	ICML(2017)	Conv4	0.13M	0.42G	50.71±0.47	58.26±0.48	62.26±0.47	64.07±0.47	65.35±0.45
RelationNet [35]	CVPR(2018)	Conv4	0.23M	0.39G	53.87±0.49	58.88±0.42	63.91±0.39	67.39±0.39	66.80±0.39
MatchingNet [34]	NIPS(2016)	Conv4	0.23M	0.40G	55.42±0.41	61.08±0.41	64.27±0.42	67.74±0.35	68.21±0.39
Protonet [36]	NIPS(2017)	Conv4	0.23M	0.40G	47.95±0.46	59.19±0.42	60.77±0.39	62.28±0.36	67.18±0.39
SPNet [10]	TGRS(2022)	Res-18	11.2M	1.82G	51.41±0.50	58.52±0.45	61.47±0.41	65.83±0.39	68.56±0.42
IDSSL [11]	GRSL(2021)	Res-18	11.2M	1.82G	57.62±0.32	67.31±0.42	71.95±0.40	74.69±0.37	76.79±0.18
HelixFormer [53]	ACM MM(2021)	Conv4	0.44M	12.87G	52.33±0.29	59.83±0.44	66.27±0.41	69.11±0.37	69.97±0.38
SetFeat [55]	CVPR(2022)	Resnet12	13.8M	4.89G	63.82±0.88	78.52±0.41	81.58±0.37	83.44±0.34	84.46±0.33
CNSPN [12]	TGRS(2022)	Res-18	13.96M	1.83G	58.16±0.48	63.50±0.45	65.94±0.41	70.09±0.40	70.31±0.40
MES^{2}L [13]	TGRS(2023)	Resnet12	8.6M	3.12G	57.71±0.47	66.05±0.43	70.09±0.39	72.65±0.39	74.6±0.37
MSFMNet [14]	TGRS(2024)	Resnet12	8.6M	3.12G	69.24±0.51	76.61±0.34	79.67±0.39	80.83±0.36	81.96±0.35
CPEA [27]	ICCV(2023)	ViT-B(CSPT)	95.8M	16.90G	55.05±0.48	65.18±0.45	70.42±0.43	72.56±0.41	75.29±0.39
SUN [24]	NIPS(2022)	Visformer	12.5M	1.00G	56.21±0.53	62.90±0.49	66.45±0.46	68.63±0.45	69.44±0.44
FEWTURE [25]	NIPS(2022)	ViT-B	86.2M	16.86G	63.70±0.48	71.00±0.41	74.96±0.37	77.42+0.34	78.49±0.33
FEWTURE [25]	NIPS(2022)	ViT-B (CSPT)	86.2M	16.86G	66.24±0.90	73.67±0.43	76.88±0.37	79.53±0.35	80.76±0.72
ProtoConViT	ours	$hybrid$	98.1M	18.68G	**76.30±0.22**	**82.06±0.37**	**85.97±0.32**	**88.01±0.29**	**90.26±0.13**
		$hybrid^*$	26.3M	5.14G	71.53±0.47	80.98±0.35	84.77±0.31	86.67±0.27	88.01±0.25
		$hybrid^\dagger$	10.5M	1.97G	71.87±0.46	79.92±0.37	83.12±0.33	85.40±0.30	86.42±0.28

The $hybrid$, $hybrid^*$ and $hybrid^\dagger$ represent Res-18+ViT-B, Res-10+ViT-S and Res-10+ViT-tiny. The best and second best results are bold and underline.

nature scene few-shot classification can also achieve good FSRSSC performance because they are dedicated to exploring spatial information existed in very limited training sample based on CNN. Besides, several FSRSSC methods (e.g., SPNet [10], IDSSL [11], CNSPN [12], MES^{2}L [13], and MSFMNet [14]) also obtain more competitive FSRSSC performance comparing with HelixFormer [53] and SetFeat [55]. Then, we can find that since MSFMNet [14] and MES^{2}L [13] individually considered the characters of remote sensing scenes to design multiscale spatial information extraction and global–local information learning ways for FSRSSC, they can get good results. In summary, whether for ViT- or CNN-based few-shot classification methods, how to effectively capture spatial contextual information is very important for FSRSSC. Moreover, in order to obviously exhibit the FSRSSC ability of the proposed ProtoConViT, the 5w1s t-SNE visualizations for 15 unseen class identifications are shown in Fig. 4; we can easily find that to compare with other methods, the proposed ProtoConViT can provide the more clear interclass decision boundary and more compact intraclass consistency under very low-data resource regimes, as shown in Fig. 4(o).

2) AID: As reported in Table II, the proposed ProtoConViT can achieve 76.30% ± 0.22%, 82.06% ± 0.37%, 85.97% ± 0.32%, 88.01% ± 0.29%, and 90.26% ± 0.13% for 5w1s, 5w2s, 5w3s, 5w4s, and 5w5s, respectively, on AID. However, the situation is different from NWPU-RESISC45, and the second best performance on AID can be achieved by the lightweight visions of hybrid architecture, e.g., $hybrid^*$ and

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TABLE III
FIVE-WAY CLASSIFICATION ACCURACIES (%) OF DIFFERENT METHODS ON THE UCM DATASET

Method	Publish	Backbone	Params	Flops	5w1s	5w2s	5w3s	5w4s	5w5s
MAML [40]	ICML(2017)	Conv4	0.13M	0.42G	46.31±0.39	58.41±0.37	60.16±0.40	63.59±0.36	64.51±0.38
RelationNet [35]	CVPR(2018)	Conv4	0.23M	0.39G	52.13±0.40	57.33±0.33	61.67±0.31	63.87±0.30	68.05±0.34
MatchingNet [34]	NIPS(2016)	Conv4	0.23M	0.40G	51.88±0.39	59.71±0.34	63.73±0.31	66.36±0.31	68.71±0.35
Protonet [36]	NIPS(2017)	Conv4	0.23M	0.40G	52.38±0.42	58.17±0.36	62.71±0.31	66.55±0.30	67.99±0.34
SPNet [10]	TGRS(2022)	Res-18	11.2M	1.82G	51.27±0.44	61.95±0.37	65.96±0.34	67.39±0.30	80.01±0.32
IDSSL [11]	GRSL(2021)	Res-18	11.2M	1.82G	52.55±0.41	61.71±0.38	66.36±0.36	69.11±0.35	70.95±0.33
HelixFormer [53]	ACM MM(2021)	Conv4	0.44M	12.87G	51.74±0.45	61.33±0.43	62.72±0.39	64.51±0.39	65.28±0.36
SetFeat [55]	CVPR(2022)	Resnet12	13.8M	4.89G	60.72±0.44	67.41±0.38	71.30±0.35	73.44±0.33	77.42±0.31
CNSPN [12]	TGRS(2022)	Res-18	13.96M	1.83G	54.62±0.50	58.80±0.47	60.61±0.42	61.94±0.38	62.24±0.38
MES^{2}L [13]	TGRS(2023)	Resnet12	12.9M	13.45G	61.21±0.43	68.49±0.4	72.92±0.35	75.64±0.33	77.25±0.32
MSFMNet [14]	TGRS(2024)	Resnet12	8.6M	3.12G	61.48±0.44	70.13±0.39	74.18±0.36	76.54±0.34	77.98±0.32
CPEA [27]	ICCV(2023)	ViT-B(CSPT)	95.8M	16.90G	63.54±0.46	72.86±0.38	76.75±0.33	78.36±0.32	81.02±0.30
SUN [24]	NIPS(2022)	Visformer	12.5M	1.00G	65.91±0.44	74.00±0.37	77.34±0.33	79.49±0.31	80.97±0.29
FEWTURE [25]	NIPS(2022)	ViT-B	86.2M	16.86G	63.65±0.45	70.49±0.41	74.88±0.38	77.72±0.35	80.10±0.33
FEWTURE [25]	NIPS(2022)	ViT-B (CSPT)	86.2M	16.86G	72.82±0.47	76.60±0.37	80.04±0.34	82.54±0.31	83.45+0.29
ProtoConViT	ours	$hybrid$	98.1M	18.68G	**79.40±0.21**	**85.58±0.32**	**86.85±0.29**	**89.68±0.24**	**91.31±0.10**
		$hybrid^*$	26.3M	5.14G	71.68±0.49	78.20±0.39	82.31±0.34	84.65±0.30	85.91±0.28
		$hybrid^\dagger$	10.5M	1.97G	72.69±0.48	79.42±0.39	82.93±0.34	84.65±0.30	85.75±0.28

The $hybrid$, $hybrid^*$ and $hybrid^\dagger$ represent Res-18+ViT-B, Res-10+ViT-S and Res-10+ViT-tiny. The best and second best results are bold and underline.

$hybrid^\dagger$ of our proposed ProtoConViT, which only require fewer Params and Flops than FewTURE [25]. This phenomenon presents that our proposed ProtoConViT has superior scalability. Moreover, as reported in Table II, to compare with other few-shot learning methods, e.g., SPNet [10], SUN [24], MatchingNet [34], RelationNet [35], ProtoNet [36], MAML [40], CPEA [27], HelixFormer [53], SetFeat [55], CNSPN [12], MES^{2}L [13], MSFMNet [14], and IDSSL [11], although different versions of the proposed ProtoConViT have no obvious advantages on the consumption of Params and Flops, a novel hybrid architecture presenting an effective classwise representation learning way according to spatial contextual clues can bring the significant performance improvement, which is very important for FSRSSC.

3) UCM Land-Use: As reported in Table III, the proposed ProtoConViT can obtain 79.40% ± 0.21%, 85.58% ± 0.32%, 86.85% ± 0.29%, 89.68% ± 0.24%, and 91.31% ± 0.10% for 5w1s, 5w2s, 5w3s, 5w4s, and 5w5s, respectively, on UCM. In addition, the lightweight versions of our proposed ProtoConViT, e.g., $hybrid^*$ and $hybrid^\dagger$, also can achieve the second best performance under most few-shot classification scenarios, such as 5w2s, 5w3s, 5w4s, and 5w5s. Then, taking into account the results in Tables I–III, comprehensively, we can see that to sufficiently explore spatial contextual information existed in very limited training samples for classwise representation is indeed beneficial for FSRSSC. Besides, since the proposed ProtoConViT can effectively integrate the advantages of CNN and ViT, whether the proposed ProtoConViT or its lightweight versions, they both exhibit stronger robustness and acceptable computational overhead.

Except to evaluate the proposed ProtoConViT on NWPU-RESISC45, AID, and UCM Land-Use based on five-way one, two, three, four, and five shots, the more challenging FSRSSC scenarios of ten-way one shot and five shots on NWPU-RESISC45 are also set up for metatraining and testing, and then, the comparison results are reported in Table IV. Next, we can find that the proposed ProtoConViT can also achieve the best performance with 84.19% ± 0.13% and 94.60% ± 0.06% accuracies for 10w1s and 10w5s. At the same time, it outperforms about 3% and 6% accuracy than the second best performance of FewTURE [25]. In addition, the model convergence speed is evaluated, as shown in Fig. 5(a) and (b). Then, we can see that under different 10w1s and 10w5s few-shot learning scenarios, the red curve lines that indicate our proposed ProtoConViT have the fast convergence speed, and it only needs about four or five training epochs to reach the highest performance for FSRSSC. For a more intuitive understanding, the confusion matrix is employed to evaluate FSRSSC performance based on 10w1s and 10w5s. As shown in Fig. 6(a)–(f), we can find that whether for 10w1s or 10w5s, the proposed ProtoConViT can perform excellent unseen class identification performance under very low-data resource regimes. Substantially, despite facing more challenging FSRSSC scenarios, the proposed ProtoConViT also can exhibit the fastest convergence speed, highest accuracy, and acceptable inference time consumption.

TABLE IV
TEN-WAY CLASSIFICATION ACCURACIES (%) OF DIFFERENT METHODS ON THE NWPU-RESISC45 DATASET

Method	Publish	Backbone	Time	10w1s	10w5s
ProtoNet [36]	NIPS(2017)	Conv4	0.02s	40.33±0.26	62.50±0.17
MatchingNet [34]	NIPS(2016)	Conv4	0.05s	43.38±0.25	58.61±0.19
SPNet [10]	TGRS(2021)	Res-18	0.09s	48.02±0.31	69.96±0.21
MSFMNet [14]	TGRS(2024)	Resnet12	0.15s	62.20±0.33	80.47±0.18
SUN [24]	NIPS(2022)	Visformer	0.32s	64.05±0.32	81.89±0.19
FEWTURE [25]	NIPS(2022)	ViT-B	1.18s	59.75±0.30	78.32±0.19
FEWTURE [25]	NIPS(2022)	ViT-B(cspt)	1.18s	78.45±0.27	91.29±0.15
ProtoConViT	ours	$hybrid$	0.45s	**84.19±0.13**	**94.60±0.06**

The $hybrid$ represents Res-18+ViT-B, and the best and second best results are presented in bold and underline, respectively.

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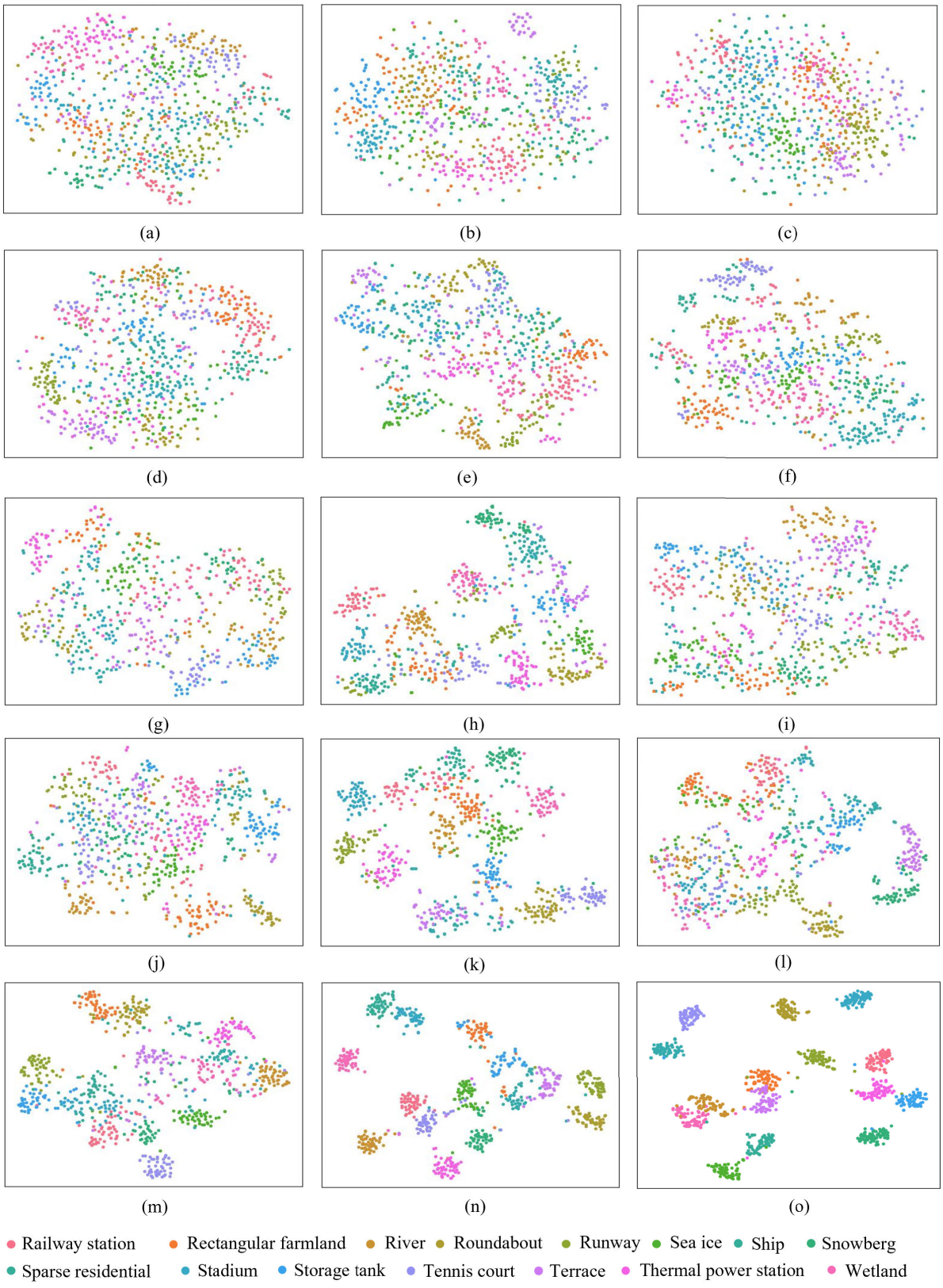


Fig. 4. t-SNE visualization of all comparison methods. The feature distributions are from 15 NWPU-RESISC45 metatesting classes with 50 randomly selected images per class for 5w1s. (a) MAML. (b) RelationNet. (c) MatchingNet. (d) ProtoNet. (e) SPNet. (f) IDSSL. (g) HelixFormer. (h) SetFeat. (i) CNSPN. (j) MES^{2}L. (k) MSFMNet. (l) CPEA. (m) SUN. (n) FEWTURE. (o) ProConViT.

D. Ablation Study and Parameter Analysis

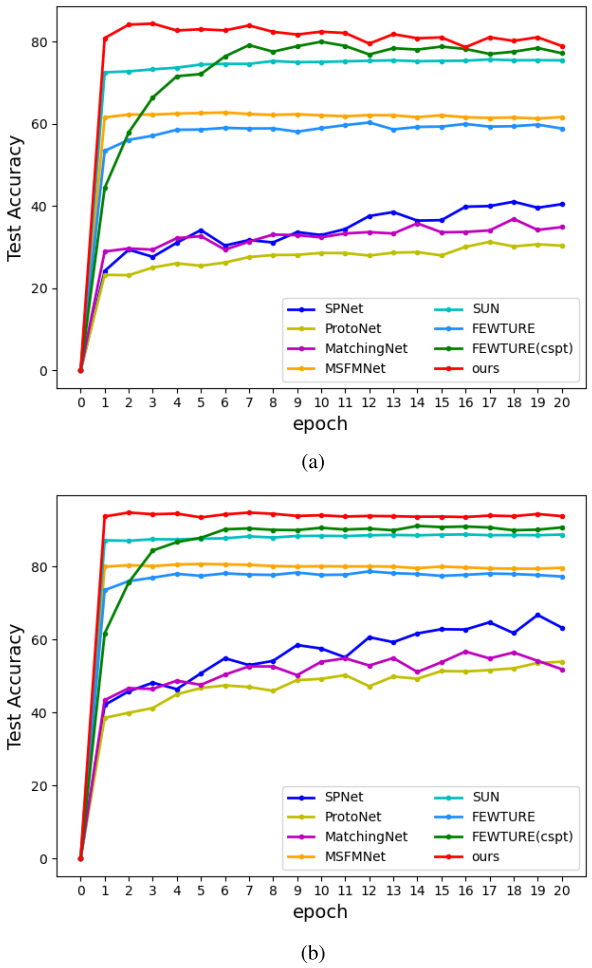
As shown in Table V, the ablation study of the proposed ProtoConViT is conducted for five-way one shot and five shots on three datasets (e.g., NWPU-RESISC45, AID, and UCM). Then, the effectiveness of different parts including the backbones, e.g., CNN and ViT, self-supervised model

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TABLE V
FIVE-WAY EXPERIMENT RESULTS (%) OF THE GENERAL ABLATION STUDY ON THE NWPU-RESISC45, AID, AND UCM LAND-USE DATASETS

Backbone		Pretrain	Hybrid Architecture		NWPU-RESISC45		AID		UCM	
Res-18	ViT-B		HPD	SQCG	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
✓					50.82±0.23	66.58±0.18	46.98±0.24	60.74±0.19	47.09±0.23	57.22±0.19
✓		✓			69.78±0.22	83.85±0.13	62.41±0.24	77.62±0.18	67.78±0.23	79.41±0.18
	✓				49.15±0.22	66.91±0.19	43.92±0.23	58.42±0.22	39.84±0.20	53.28±0.17
	✓	✓			86.63±0.17	95.06±0.08	68.69±0.25	81.72±0.17	72.32±0.23	83.67±0.16
✓	✓	✓	✓		90.51±0.15	96.84±0.06	74.85±0.22	89.15±0.17	78.16±0.22	91.24±0.11
✓	✓	✓	✓	✓	91.04±0.15	97.32±0.06	76.30±0.22	90.26±0.13	79.40±0.21	91.31±0.10

Pretrain: self-supervised model pretraining; HPD: heterogeneous prototype distillation; SQCG: support-query correlative guidance



(a)

(b)

Fig. 5. Ten-way metatesting accuracy on the NWPU-RESISC45 dataset. (a) 10w1s. (b) 10w5s.

pretraining, and hybrid architecture, would be discussed as follows.

1) Backbones: According to the first and third lines in Table V, when the naive CNN (i.e., Res-18) and plain ViT (i.e., ViT-B) are directly used for FSRSSC according to ProtoNet [36], we can find that they both provide poor FSRSSC performance under five-way one shot and five shots on NWPU-RESISC45, AID, and UCM. Yet, the naive CNN can provide slightly better results than the plain ViT because the inductive bias of CNN can be seen as the prior information, which is beneficial for few-shot learning to a certain extent. Meanwhile, massive learnable parameters with redundant representation space of plain ViT would obstruct to search for critical contextual clues from very limited training samples; thus, plain ViT has a big risk of overfitting and leading to performance collapse for FSRSSC.

2) Self-Supervised Model Pretraining: To compare with the first and third lines, the second and fourth lines in Table V show that when CNN and ViT undergo the self-supervised model pretraining in-domain, their FSRSSC performances have tremendous improvement. This phenomenon has demonstrated that the self-supervised model pretraining according to unlabeled data in-domain can effectively unleash the huge potential of few-shot learning ability whether for CNN or ViT. Thus, the self-supervised model pretraining can be seen as an essential step to set up a generalized class-agnostic representation for metalearning, which is similar to HVCS that has the few-shot classification ability for unseen classes through long-term learning and valid knowledge accumulation. Especially for ViT, we can see that after self-supervised model pretraining, the ViT-B can obtain amazing FSRSSC results because if the few-shot learning ability of ViT is unleashed, the much more consistent spatial contextual clues between unseen and seen classes can be rapidly captured for classwise representation, which would possess the huge potential adapting to agnostic biases (i.e., data or classes) from diverse metatesting scenarios.

3) Hybrid Architecture: As reported in the fifth and sixth lines of Table V, when discriminative multimodality information of CNN is gracefully integrated into transferable spatial contextual clues from ViT, it would provide superior FSRSSC performance. Specifically, individually comparing the second and fourth lines with the fifth line in Table V, we can find that the designed heterogeneous prototype distillation can effectively integrate the advantages of CNN and ViT to emerge a more powerful few-shot learning ability because much more critical spatial contextual clues can be rapidly captured from very limited training samples, and transferable and discriminative concepts would be formed by heterogeneous prototype representation. Thus, it can largely improve FSRSSC performance for 5w1s and 5w5s than by singly using the pretrained CNN or ViT on NWPU-RESISC45, AID, and

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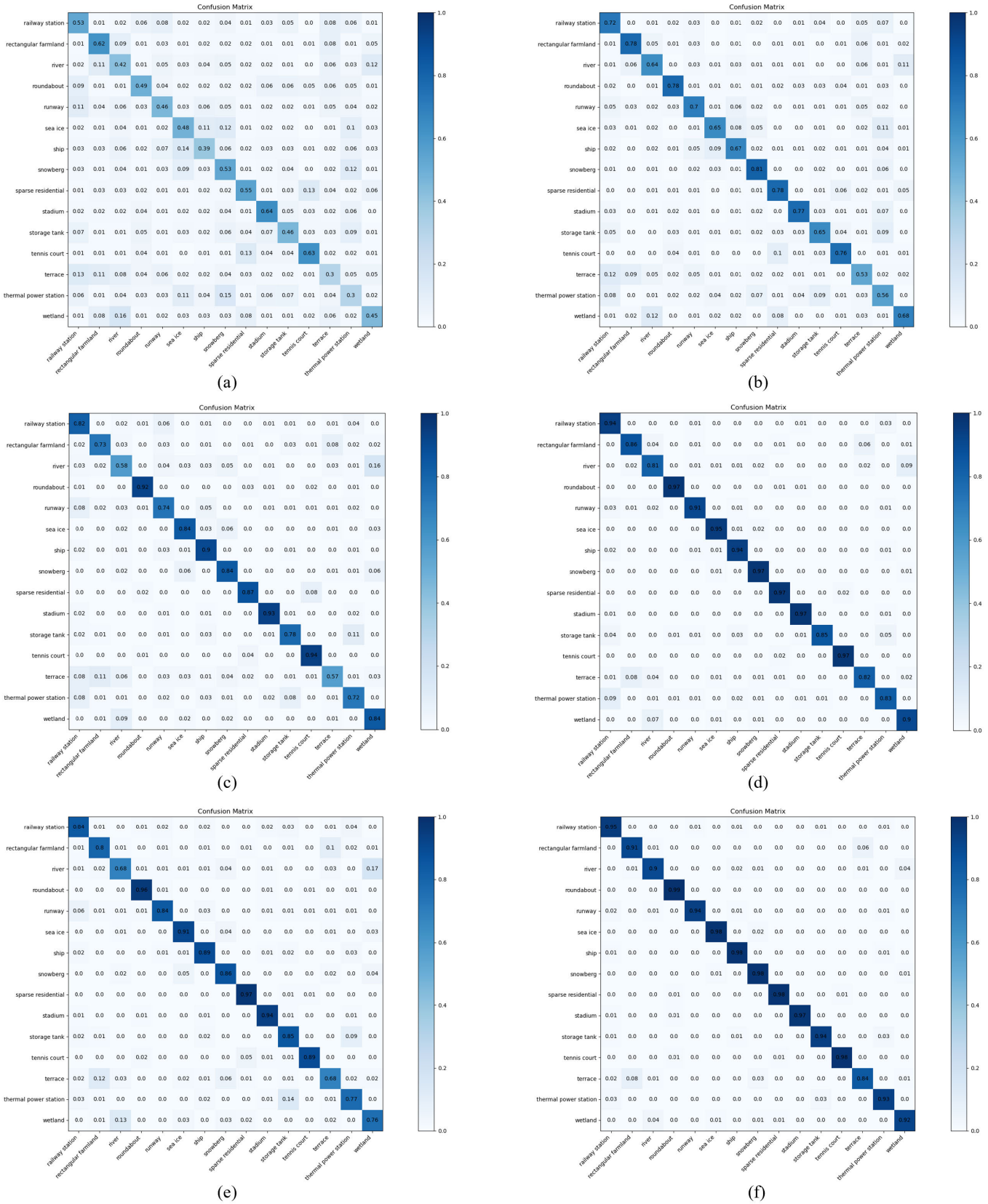


Fig. 6. Confusion matrix of comparison methods on NWPU-RESISC45. (a) SPNet for 10w1s. (b) SPNet for 10w5s. (c) FEWTURE(CSPT) for 10w1s. (d) FEWTURE(CSPT) for 10w5s. (e) Proposed ProtoConViT for 10w1s. (f) Proposed ProtoConViT for 10w5s.

UCM. On the other hand, this phenomenon also indicates that the inductive bias of CNN is injected into heterogeneous prototype representation, which can help ViT rapidly learn the critical spatial contextual clues and, therefore, perfectly

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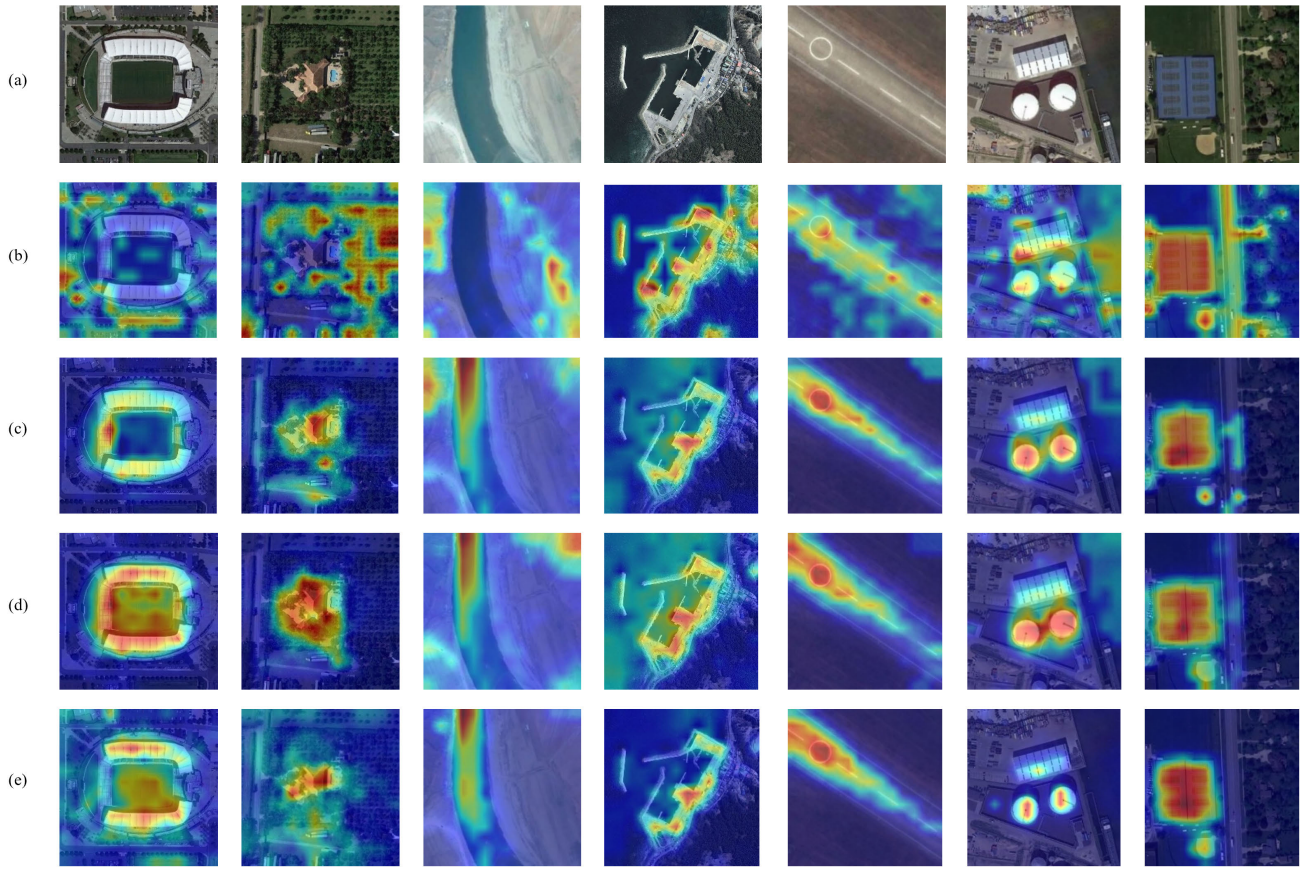
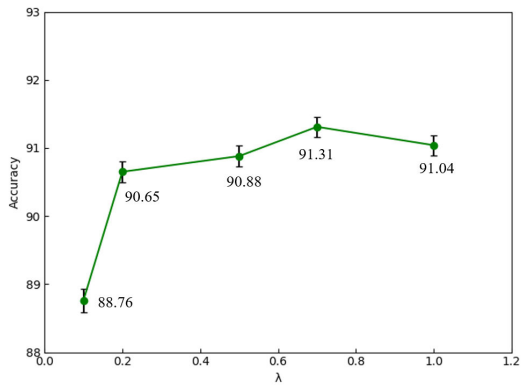


Fig. 7. Attention map analysis on NWPU-RESISC45. (a) Original remote sensing scene images. (b) Score maps produced by ViT without pretraining. (c) Score maps produced by pretrained ViT. (d) Score maps produced by the designed heterogeneous prototype distillation without support-query correlative guidance. (e) Exhibits score maps produced by the proposed ProtoConViT.

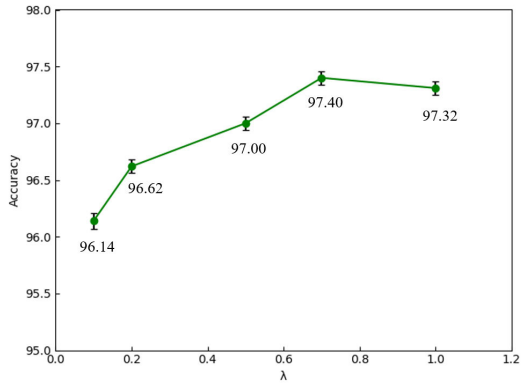
fit for few-shot learning scenarios. Finally, in order to reduce the redundant spatial contextual representation learning space of ViT and then further improve the few-shot learning ability for heterogeneous prototype representation, support-query correlative guidance is proposed for heterogeneous prototype distillation. To compare the fifth line with the sixth line in Table V, we can see that the FSRSSC performance is further improved for 5w1s and 5w5s on NWPU-RESISC45, AID, and UCM, which indicates that the designed support-query correlative guidance can effectively assist heterogeneous prototype representation in rapidly setting up discriminative and transferable concepts among support and query data, within each episodic training task. Next, for obviously exhibiting the few-shot learning abilities of different representation learning ways for capturing the critical spatial contextual clues, several attention maps are shown in Fig. 7. Referring to original remote sensing scenes in Fig. 7(a), we can find that when ViT without undergoing any model pretraining as shown in Fig. 7(b), it would randomly capture spatial contextual clues from redundant spatial information and result in inferior FSRSSC performance. Then, as shown in Fig 7(c), after model pretraining, ViT indeed has a few-shot learning ability to a certain extent, which begins to pay close attention to meaningful spatial contextual clues from very limited training samples. Subsequently, from Fig. 7(d), we can see that the more critical contextual clues can be learned by heterogeneous prototype distillation for FSRSSC because the inductive of CNN can be seamlessly integrated into heterogeneous prototype representation to improve few-shot learning ability and form the transferable and discriminative concept for FSRSSC. In the end, as shown in Fig. 7(e), the spatial contextual representation learning space would be reduced by the designed support-query correlative guidance. Thereby, the few-shot learning ability of the proposed ProtoConViT can be improved, and it can rapidly set up the more discriminative and transferable concepts from very limited training samples and then be used for unseen class identification.

4) Parameter Analysis: After the ablation study, the impact of hyperparameter λ for balancing discriminative multimodality information integrated into transferable spatial contextual clues for heterogeneous prototype representation should be analyzed. As mentioned in Section III-C, corresponding to the proposed ProtoConViT, the joint loss of (9) contains two terms L_{vit} and L_{cnn} for heterogeneous prototype distillation L_{KD}. Here, as shown in the analysis in Table V and Fig. 7, we can clearly know that under the designed support-query correlative guidance, heterogeneous prototype distillation can seamlessly introduce inductive bias of CNN into ViT to rapidly set up heterogeneous prototype representation. Subsequently, the inductive bias of CNN can encourage ViT to capture much more critical contextual clues among support and query data and, meanwhile, form transferable and discriminative concepts for FSRSSC. Therefore, how CNN branch L_{cnn} to affect the performance of heterogeneous prototype representation

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(a)



(b)

Fig. 8. Discussion of parameter λ on NWPU-RESISC45. (a) 5w1s. (b) 5w5s.

is discussed in Fig. 8(a) and (b). We can find that when λ becomes larger toward 1, the FSRSSC performances for 5w1s and 5w5s are gradually improved. When $\lambda = 0.7$, the collaborative effect between ViT and CNN is optimal, which can achieve 91.31% accuracy for 5w1s and 97.40% accuracies for 5w5s on NWPU-RESISC45. As a result, the hyperparameter λ should be defined as 0.7 to effectively trade off the transferability and discriminability for heterogeneous prototype representation.

V. Conclusion

In this article, aiming to explore abundant spatial contextual information that existed in very limited training samples for classwise representation, finely adapting to agnostic biases in diverse metatesting scenarios, a novel hybrid architecture called ProtoConViT consisting of CNN and ViT is proposed for FSRSSC. Initially, the class-agnostic generalized heterogeneous knowledge is acquired through self-supervised model pretraining, which can not only encourage a few-shot learning ability but also prevent the overfitting phenomenon. Subsequently, a heterogeneous prototype distillation with support-query spatial correlative guidance is designed to effectively integrate discriminative multimodality information into transferable spatial contextual clues, which not only can make the model intend to capture much more critical spatial contextual clues within support-query correlative spatial regions but also can facilitate forming transferable and discriminative concepts by heterogeneous prototype construction. Finally, extensive experiments have been carried out to prove the effectiveness of the proposed ProtoConViT, which can achieve the SOTA FSRSSC performance. In our future work, we will continuously focus on exploring how to build an effective representation with even fewer parameters and low computational requirements under a very low-data resource regime. In addition, except for the exploration of representation spaces, more rational decision modes are also crucial for FSRSSC.

References

[1] Z. Sha and J. Li, "MITformer: A multiinstance vision transformer for remote sensing scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[2] X. Tang, M. Li, J. Ma, X. Zhang, F. Liu, and L. Jiao, "EMTCAL: Efficient multiscale transformer and cross-level attention learning for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5626915.

[3] P. Lv, W. Wu, Y. Zhong, F. Du, and L. Zhang, "SCViT: A spatial-channel feature preserving vision transformer for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4409512.

[4] J. Chen, J. Yi, A. Chen, and Z. Jin, "EFCOMFF-Net: A multiscale feature fusion architecture with enhanced feature correlation for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5604917.

[5] G. Wang et al., "P2FEViT: Plug-and-play CNN feature embedded hybrid vision transformer for remote sensing image classification," *Remote Sens.*, vol. 15, no. 7, p. 1773, Mar. 2023.

[6] D. Derksen, G. Meoni, G. Lecuyer, A. Mergy, M. Märtens, and D. Izzo, "Few-shot image classification challenge on-board," in *Proc. Workshop-Data Centric AI, NeurIPS*, 2021.

[7] L. Wang, X. Bai, C. Gong, and F. Zhou, "Hybrid inference network for few-shot SAR automatic target recognition," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9257–9269, Nov. 2021.

[8] T. Zhao, M. H. Cosh, A. Roy, X. Mu, Y. Qiu, and J. Shi, "Remote sensing experiments for earth system science," *Int. J. Digit. Earth*, vol. 14, no. 10, pp. 1237–1242, Oct. 2021.

[9] G. Caspari, S. Donato, and M. Jendryke, "Remote sensing and citizen science for assessing land use change in the Musandam (Oman)," *J. Arid Environ.*, vol. 171, Dec. 2019, Art. no. 104003.

[10] G. Cheng et al., "SPNet: Siamese-prototype network for few-shot remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5608011.

[11] C. Tao, J. Qi, W. Lu, H. Wang, and H. Li, "Remote sensing image scene classification with self-supervised paradigm under limited labeled samples," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[12] J. Chen et al., "Improving few-shot remote sensing scene classification with class name semantics," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5633712.

[13] J. Li, M. Gong, H. Liu, Y. Zhang, M. Zhang, and Y. Wu, "Multi-form ensemble self-supervised learning for few-shot remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4500416.

[14] S. Wang et al., "Personalized multiparty few-shot learning for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4506115.

[15] C. Doersch, A. Gupta, and A. Zisserman, "CrossTransformers: Spatially-aware few-shot transfer," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 21981–21993.

[16] J. Wu, T. Zhang, Y. Zhang, and F. Wu, "Task-aware part mining network for few-shot learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2021, pp. 8433–8442.

[17] C. Dong, W. Li, J. Huo, Z. Gu, and Y. Gao, "Learning task-aware local representations for few-shot learning," in *Proc. Int. Joint Conf. Artif. Intell. (IJCAI)*, Aug. 2021, pp. 716–722.

[18] B. Jiang, K. Zhao, and J. Tang, "RGTransformer: Region-graph transformer for image representation and few-shot classification," *IEEE Signal Process. Lett.*, vol. 29, pp. 792–796, 2022.

[19] B. Zhang et al., "SGMNet: Scene graph matching network for few-shot remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5628915.

[20] Z. Yuan, C. Tang, A. Yang, W. Huang, and W. Chen, "Few-shot remote sensing image scene classification based on metric learning and local descriptors," *Remote Sens.*, vol. 15, no. 3, p. 831, Feb. 2023.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:47:07 UTC from IEEE Xplore. Restrictions apply.

[21] J. He, A. Kortylewski, and A. Yuille, "CORL: Compositional representation learning for few-shot classification," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2023, pp. 3879–3888.

[22] S. Jiang, Y. Zhu, C. Liu, X. Song, X. Li, and W. Min, "Dataset bias in few-shot image recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 229–246, Jan. 2023.

[23] Z. Lu, H. Xie, C. Liu, and Y. Zhang, "Bridging the gap between vision transformers and convolutional neural networks on small datasets," in *Proc. 35st Conf. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 14663–14677.

[24] B. Dong, P. Zhou, S. Yan, and W. Zuo, "Self-promoted supervision for few-shot transformer," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2022, pp. 329–347.

[25] M. Hiller, R. Ma, M. Harandi, and T. Drummond, "Rethinking generalization in few-shot classification," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, vol. 35, 2022, pp. 3582–3595.

[26] C. Xu, S. Yang, Y. Wang, Z. Wang, Y. Fu, and X. Xue, "Exploring efficient few-shot adaptation for vision transformers," 2023, *arXiv:2301.02419*.

[27] F. Hao, F. He, L. Liu, F. Wu, D. Tao, and J. Cheng, "Class-aware patch embedding adaptation for few-shot image classification," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 18905–18915.

[28] M. M. Naseer et al., "Intriguing properties of vision transformers," in *Proc. NIPS*, vol. 34, 2021, pp. 23296–23308.

[29] D. Wang, J. Zhang, B. Du, G.-S. Xia, and D. Tao, "An empirical study of remote sensing pretraining," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5608020.

[30] G.-S. Xia et al., "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017.

[31] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017.

[32] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proc. 18th SIGSPATIAL Int. Conf. Adv. Geograph. Inf. Syst.*, Nov. 2010, pp. 270–279.

[33] Y. Song, T. Wang, P. Cai, S. K. Mondal, and J. P. Sahoo, "A comprehensive survey of few-shot learning: Evolution, applications, challenges, and opportunities," *ACM Comput. Surv.*, vol. 55, no. 13s, pp. 1–40, Dec. 2023.

[34] O. Vinyals, C. Blundell, T. Lillicrap, K. Kavukcuoglu, and D. Wierstra, "Matching networks for one shot learning," in *Advances in Neural Information Processing Systems*, vol. 29. Red Hook, NY, USA: Curran Associates, 2016.

[35] F. Sung, Y. Yang, L. Zhang, T. Xiang, P. H. S. Torr, and T. M. Hospedales, "Learning to compare: Relation network for few-shot learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Salt Lake City, UT, USA, Jun. 2018, pp. 1199–1208.

[36] J. Snell, K. Swersky, and R. Zemel, "Prototypical networks for few-shot learning," in *Advances in Neural Information Processing Systems*, vol. 30. Red Hook, NY, USA: Curran Associates, 2017.

[37] H. Cheng, Y. Wang, H. Li, A. C. Kot, and B. Wen, "Disentangled feature representation for few-shot image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, pp. 1–14, 2023.

[38] J. Xie, F. Long, J. Lv, Q. Wang, and P. Li, "Joint distribution matters: Deep Brownian distance covariance for few-shot classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 7972–7981.

[39] A. Singh and H. Jamali-Rad, "Transductive decoupled variational inference for few-shot classification," 2022, *arXiv:2208.10559*.

[40] C. Finn, P. Abbeel, and S. Levine, "Model-agnostic meta-learning for fast adaptation of deep networks," in *Proc. 34th Int. Conf. Mach. Learn.*, vol. 70, Aug. 2017, pp. 1126–1135.

[41] A. Nichol, J. Achiam, and J. Schulman, "On first-order meta-learning algorithms," 2018, *arXiv:1803.02999*.

[42] A. Antoniou, H. Edwards, and A. Storkey, "How to train your MAML," 2018, *arXiv:1810.09502*.

[43] D. S. Pandey and Q. Yu, "Multidimensional belief quantification for label-efficient meta-learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 14371–14380.

[44] H. Yao, L. Zhang, and C. Finn, "Meta-learning with fewer tasks through task interpolation," 2021, *arXiv:2106.02695*.

[45] M. Yazdanpanah et al., "Revisiting learnable affines for batch norm in few-shot transfer learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 9109–9118.

[46] W.-H. Li, X. Liu, and H. Bilen, "Universal representation learning from multiple domains for few-shot classification," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9506–9515.

[47] A. Chowdhury, M. Jiang, S. Chaudhuri, and C. Jermaine, "Few-shot image classification: Just use a library of pre-trained feature extractors and a simple classifier," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9425–9434.

[48] K. Chen and C.-G. Lee, "Meta-free few-shot learning via representation learning with weight averaging," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jul. 2022, pp. 1–8.

[49] Y. Chen, Z. Liu, H. Xu, T. Darrell, and X. Wang, "Meta-baseline: Exploring simple meta-learning for few-shot learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9062–9071.

[50] S. X. Hu, D. Li, J. Stühmer, M. Kim, and T. M. Hospedales, "Pushing the limits of simple pipelines for few-shot learning: External data and fine-tuning make a difference," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 9068–9077.

[51] P. Chikontwe, S. Kim, and S. H. Park, "CAD: Co-adapting discriminative features for improved few-shot classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 14534–14543.

[52] S. Lee, W. Moon, and J.-P. Heo, "Task discrepancy maximization for fine-grained few-shot classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 5321–5330.

[53] B. Zhang, J. Yuan, B. Li, T. Chen, J. Fan, and B. Shi, "Learning cross-image object semantic relation in transformer for few-shot fine-grained image classification," in *Proc. 30th ACM Int. Conf. Multimedia*, Oct. 2022, pp. 2135–2144.

[54] C. Cao and Y. Zhang, "Learning to compare relation: Semantic alignment for few-shot learning," *IEEE Trans. Image Process.*, vol. 31, pp. 1462–1474, 2022.

[55] A. Afrasiyabi, H. Larochelle, J.-F. Lalonde, and C. Gagné, "Matching feature sets for few-shot image classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 9014–9024.

[56] P. Gao, J. Lu, H. Li, R. Mottaghi, and A. Kembhavi, "Container: Context aggregation network," 2021, *arXiv:2106.01401*.

[57] Z. Peng et al., "Conformer: Local features coupling global representations for visual recognition," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 367–376.

[58] Y. Xie and Y. Liao, "Efficient-ViT: A light-weight classification model based on CNN and ViT," in *Proc. 6th Int. Conf. Image Graph. Process.*, Jan. 2023, pp. 64–70.

[59] P. Gao, T. Ma, H. Li, Z. Lin, J. Dai, and Y. Qiao, "ConvMAE: Masked convolution meets masked autoencoders," 2022, *arXiv:2205.03892*.

[60] H. Zhang, J. Duan, M. Xue, J. Song, L. Sun, and M. Song, "Bootstrapping ViTs: Towards liberating vision transformers from pre-training," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 8944–8953.

[61] P. Deng, K. Xu, and H. Huang, "When CNNs meet vision transformer: A joint framework for remote sensing scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[62] K. Xu, P. Deng, and H. Huang, "Vision transformer: An excellent teacher for guiding small networks in remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5618715.

[63] K. Li et al., "UniFormer: Unifying convolution and self-attention for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 10, pp. 12581–12600, Oct. 2023.

[64] S. Gidaris, A. Bursuc, N. Komodakis, P. Pérez, and M. Cord, "Boosting few-shot visual learning with self-supervision," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2019, pp. 8059–8068.

[65] C. Perng Phoo and B. Hariharan, "Self-training for few-shot transfer across extreme task differences," 2020, *arXiv:2010.07734*.

[66] V. Risojević and V. Stojnić, "Do we still need ImageNet pre-training in remote sensing scene classification?" 2021, *arXiv:2111.03690*.

[67] T. Zhang et al., "Consecutive pre-training: A knowledge transfer learning strategy with relevant unlabeled data for remote sensing domain," *Remote Sens.*, vol. 14, no. 22, p. 5675, Nov. 2022.

[68] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9729–9738.

[69] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 16000–16009.

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[70] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[71] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.



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